Missing Mass and Minimum Wages: Distributional Changes around Four Minimum Wage Increases in Peru

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Abstract

When a country raises its minimum wage, what happens to the workers who were paid less than the new floor? Peru raised its floor four times between 2016 and 2025, a cumulative 51 percent. Its floor is national, so I compare each wage distribution before and after the increase against a modelled counterfactual. Three household surveys, two of them quarterly, give the same answer. The band of wages below the new floor empties and the band just above it fills, under every counterfactual and in every survey. In the best-measured increase the band below loses 5.0 to 5.2 percentage points while the band just above gains 2.5 to 2.7, and the rest of the wage distribution accounts for the difference. The quarterly data date the movement to within three quarters. Pay concentrates at the new floor, and the range beneath it thins by more than the range above it fills.

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1 Introduction

Peru raised its minimum wage (MW, *Remuneración Mínima Vital* or RMV) four times between 2016 and 2025, from S/750 to S/1,130, a cumulative 51% nominal increase. What happened to the wage distribution? Did workers near the minimum wage floor receive raises, or did they lose their jobs? Peru's labor market has high informality. Under the comprehensive definition of the Instituto Nacional de Estadística e Informática (INEI), informal employment accounts for roughly 70 to 75 percent of total employment across the period studied here (Instituto Nacional de Estadística e Informática, 2024), while the RMV applies only to registered formal dependent workers.

Peru has a single national MW with no sub-national variation in the policy. Cengiz et al. (2019) construct their counterfactual by comparing U.S. states with MW changes to states without, a difference-in-differences design. This is not feasible in Peru. Instead, I follow the methodological precedent of Harasztosi and Lindner (2019), who analyze Hungary's national MW using the pre-reform wage distribution as the counterfactual, with a background correction from the upper tail of the distribution.

This paper uses a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for Peru's national MW setting, to answer these questions. The estimator compares an observed wage distribution against a counterfactual and reports two signed quantities: the change in the share of workers below the new floor, and the change in the share just above it. Each increase leaves a band of workers earning below a floor that has just moved above them, and the two quantities ask what became of that band: how much of it emptied out, and how much of it reappeared immediately above the new minimum. The population is formal dependent workers in the annual national data and all dependent workers in the two quarterly series, and every display states which. These are the paper's estimands, defined formally in Section 4. With the change in the remainder of the distribution they sum to zero by construction.

I analyze four MW events: Event A (S/750 to S/850, May 2016), Event B (S/850 to

S/930, April 2018), Event C (S/930 to S/1,025, May 2022) and Event D (S/1,025 to S/1,130, January 2025). For Events A, B and D the distribution moves as a wage floor predicts. Mass leaves the depletion zone below the new minimum and accumulates in the zone just above it, with the share above the floor rising and the share below it falling under every counterfactual baseline in every survey that observes the event. In the annual national data the accumulation-zone gain runs from 2.1 to 5.4 percentage points across baselines and events, in Lima from 2.9 to 4.4, and in the national quarterly series for Event D from 2.5 to 2.7. The two do not offset. The band below the floor loses more than the band just above it gains, in every event, under every baseline and in every survey, and the rest of the wage distribution accounts for the difference. Event C is not comparable to the others and is reported as descriptive evidence only.

Three counterfactual baselines agree, no other date reproduces the pattern, and the statutory threshold dominates every artificial one, so the movement is concentrated around the statutory floor. The department-level regressions of Section 4.4 rest on a separate identifying assumption.

The bunching ratio R, which much of this literature reports, appears here as a descriptive accounting statistic, and the signed changes carry the inference throughout.

The closest antecedents are of two kinds. Cengiz et al. (2019) and Harasztosi and Lindner (2019) supply the distributional method, the first with cross-state variation and the second with a national floor and a pre-reform counterfactual. For Peru, Jaramillo Baanante (2004) finds formal-sector earnings effects confined to workers caught between the old and the new minima, Céspedes (2006) documents a perceptible change in the formal Lima wage distribution around the floor, Céspedes and Sánchez (2014) estimate employment responses on the EPE Lima series that are larger in small firms, Jaramillo Baanante (2012) finds substitution toward more skilled workers and no contraction, and Corcuera García (2024) finds reallocation towards self-employment alongside negative employment elasticities larger than the recent literature. This paper differs from all of them in reporting signed distributional

changes with survey-design inference, and in adding a high-frequency Lima series that dates the adjustment.

This paper makes four contributions. First, it measures Peru's minimum wage increases with signed distributional estimands in place of a ratio. Both signed changes are reported against three counterfactual baselines with survey-design inference. The placebo exercises assess the specificity of the accumulation signal and the joint gross-mass pattern (relating to Harasztosi and Lindner, 2019; Saltiel and Urzúa, 2022; Gindling and Terrell, 2009; Engbom and Moser, 2022; Corcuera García, 2024). Second, it pairs national breadth with high-frequency timing, and for the 2025 increase it delivers both at once. The rolling three-month Lima series shows that the 2018 adjustment appears in the first fully post-policy window and holds, whereas the 2016 adjustment builds over roughly a further quarter. The national quarterly series measures the 2025 increase against ten pre-policy quarters, with no window mixing treated and untreated months, and its estimates are the tightest in the paper. Third, it identifies a measurement sensitivity of discrete-bin distributional estimators to where the policy threshold falls inside a bin. Misalignment at S/50 bins materially distorts the measured zone masses, while the 2025 increase shows that a small misalignment moves them by 0.01 percentage points, so the distortion scales with the size and direction of the offset. Fourth, the department-level analysis documents an episode-specific composition gradient. Following the 2018 increase, a 10-percentage-point increase in pre-policy exposure is associated with a 4.0 to 4.3 percentage-point larger increase in informal self-employment, including after conditioning on exposure to the 2016 increase. Employment and formal-employment estimates show no systematic post-policy pattern in the two pre-pandemic episodes (Card and Krueger, 1994; Neumark and Shirley, 2022; Manning, 2021; Dube, 2019; Belman and Wolfson, 2014). The composition gradient is attenuated when Lima is excluded and is not present in the other episodes.

The paper proceeds as follows. Section 2 describes Peru's minimum wage institutions and the four events, and Section 3 the three surveys. Section 4 sets out the distributional

estimands, the counterfactual baselines and the inference. Section 5 reports the signed distributional results for all four events and dates the adjustment. Section 6 places them against the Peruvian evidence, Section 7 reports the composition and department-level evidence, and Section 8 the placebo and robustness tests. Section 9 concludes. Hours, subgroup comparisons, compression and the non-identified employment designs are in the appendix.

2 Institutional Setting and the Four Increases

2.1 Peru's Minimum Wage System

The *Remuneración Mínima Vital* (RMV) is set by executive decree and applies to all registered formal dependent workers, which for most of the period studied here means *empleados* and *obreros*, with *trabajadores del hogar* entering the entitlement in November 2020 as the next paragraph sets out. It is not indexed to inflation or productivity and changes only when the government issues a new decree. Enforcement is conducted by SUNAFIL (*Superintendencia Nacional de Fiscalización Laboral*), whose inspections are concentrated in the more formal departments, Lima above all.

Two features of that entitlement bear on the distributional results. First, the monthly figure corresponds to a full working day of eight hours, or forty-eight hours a week, and shorter schedules are remunerated in proportion, so a monthly wage below the floor is lawful for a worker on a reduced *jornada*. The analysis sample imposes no hours restriction, and in the national quarterly series 45.8 percent of dependent workers earning below S/1,130 report fewer than 35 usual weekly hours. Second, *trabajadores del hogar* are included in the ENAHO dependent-worker definition throughout, but they were brought into the private-sector regime, and so into the RMV entitlement, only by Ley 31047 of November 2020. For the 2016 and 2018 increases they were therefore outside the entitlement while inside the sample. They are a small group: 0.9 percent of the formal dependent sample and 2.8 to 3.0 percent of the affected zone, defined in Section 4.1, in those two pre-periods.

Records below the monthly threshold are observations located below a distributional cut-off, and the zone labels used throughout are positional.

The structure of Peru's informal sector, and the policy attempts to reduce it, are treated in Yamada (1996), Chacaltana and Sulmont (2004) and Díaz (2014). The RMV has been studied as a labor market anchor and its effects on wages, employment, and formality have been documented across multiple reform episodes (Jaramillo Baanante, 2004; Céspedes, 2006; Céspedes and Sánchez, 2014; Jaramillo Baanante, 2012; Corcuera García, 2024).

2.2 The Four MW Events

Table 1 summarizes the four MW changes analyzed in this paper. The first three are separated by approximately two years or more, allowing clean pre-post comparisons using annual ENAHO surveys. The fourth falls outside the ENAHO span used here and is observed only in the national high-frequency series, where it has ten pre-policy quarters.

Table 1: Minimum Wage Events Analyzed

Event	Pre-MW	Post-MW	Change	Effective	Analysis periods
A	S/750	S/850	+13.3%	May 2016	Pre: 2015, Post: 2017
B	S/850	S/930	+9.4%	April 2018	Pre: 2017, Post: 2019
C	S/930	S/1,025	+10.2%	May 2022	Pre: 2021, Post: 2023
D	S/1,025	S/1,130	+10.2%	January 2025	Pre: 2022Q3–2024Q4, Post: 2025Q1–2026Q1

Note: The 2020 ENAHO is excluded because COVID-19 methodological modifications and reduced sample size limit comparability. Event C's pre-period (2021) reflects post-pandemic labor market conditions. Events A to C are measured in the annual national survey and in the Lima high-frequency series. Event D falls outside the annual span used here and is measured only in the national high-frequency series, in calendar quarters. Its effective date falls on a quarter boundary, so no window mixes pre-policy and post-policy months. Statutory sources: DS 005-2016-TR, DS 004-2018-TR, DS 003-2022-TR and DS 006-2024-TR.

3 Data

3.1 ENAHO National Annual Data

3.1.1 ENAHO Module 500

The primary data source is the *Encuesta Nacional de Hogares* (ENAHO) Module 500 (*Empleo e Ingresos*), conducted annually by INEI. The main analysis uses cross-sectional data from eight years: 2015–2019, 2021–2023 (2020 is excluded due to COVID-19 methodological modifications). ENAHO 2013 and 2014 enter as auxiliary data in two places. They anchor the pre-policy placebo audit in Section 7.2, in which the share-of-treated DiD is applied to a window with no minimum wage change as a diagnostic on the identifying assumption, and they are part of the training window for the two model-based counterfactual baselines of Section 4.2.

The analysis sample consists of formal dependent workers (*empleados*, *obreros*, and *trabajadores del hogar*) under INEI's formality definition, with monthly cash income from the primary occupation between S/1 and S/6,000 and positive survey weights. Throughout, the canonical row-level wage rule takes *p524a1*, monthly cash income from the primary occupation, where that variable is positive, and otherwise substitutes *i524a1* divided by twelve for that same record. The substitution is made row by row and never for whole years, which is what "row-level" denotes. Department is identified from the first two digits of the geographic identifier. The exact ENAHO Module 500 variables used and their definitions appear in Appendix O.

3.1.2 Summary Statistics

Table 2 presents summary statistics for the analysis sample across all eight years.

Table 2: Summary Statistics: Formal Dependent Workers, ENAHO 2015–2023

	2015	2016	2017	2018	2019	2021	2022	2023
MW (S/.)	750	850	850	930	930	930	1,025	1,025
N	10,059	11,488	10,895	11,723	10,922	8,946	9,554	9,838
Median wage (wt.)	1,300	1,500	1,500	1,500	1,500	1,500	1,600	1,700
Median wage (unwt.)	1,400	1,500	1,500	1,673	1,700	1,800	1,800	1,863
% at MW	5.1	5.1	6.9	5.9	7.7	10.5	3.1	7.1
% below MW	11.6	14.5	10.4	15.1	12.1	11.6	17.8	12.7
Median Kaitz	0.577	0.567	0.567	0.620	0.620	0.620	0.641	0.603
Mean hrs/wk	46.4	46.2	46.4	46.4	46.2	47.0	46.7	46.6
Formality (%)	26.7	27.9	27.4	27.6	27.3	23.2	24.3	26.1

Note: N = unweighted formal dependent workers with monthly wage S/1–6,000 and positive survey weight. % at MW = share in the S/25 bin beginning at the MW, $[\text{MW}, \text{MW} + 25)$. % below MW = share earning below MW. Median Kaitz = MW / median wage, national. Formality rate = formal dependent / all employed. The 2020 ENAHO is excluded due to reduced sample size and methodological modifications during COVID-19, which limit comparability with other years. COVID-19 year 2021 shows depressed formality reflecting pandemic-period job losses.

The weighted median rises from S/1,300 (2015) to S/1,700 (2023), a 30.8 percent nominal increase, broadly tracking the 36.7 percent increase in the statutory minimum wage. The weighted median persists at S/1,500 from 2016 through 2021 because round-number reporting concentrates between 6.2 and 8.5 percent of total formal dependent worker weight on exactly S/1,500 in each of those years, creating a plateau that the survey weights amplify but that does not reflect a true stagnation in central-tendency formal wages. The unweighted median rises monotonically from S/1,400 (2015) to S/1,863 (2023), reported alongside the weighted median in Table 2 to make the heaping artifact transparent. Both series tell the same qualitative story of formal wage growth tracking the MW. The Kaitz index (weighted)

ranges from 0.567 to 0.641 across the eight years, rising as the MW increases relative to the median formal wage. The share below the contemporaneous MW is elevated in the policy-change years (14.5% in 2016, 15.1% in 2018, 17.8% in 2022) and then falls.[1] Weekly hours are stable at 46–47 hours throughout the period.

3.2 EPE Lima Rolling Three-Month Data

EPE publishes rolling three-month windows on a monthly schedule, so consecutive releases share two of their three interview months by construction. The expansion factor is labelled in the data dictionary as a *Factor de Expansión Trimestral*, and the technical documentation names the moving quarter as the only temporal level of inference. Monthly estimates are not supported, and none are reported. The unit of observation is the rolling window, and for each window I record how many of its three interview months fall after the increase. A three-month window crossing a policy date always produces exactly two partially treated windows, so no sharp monthly break is identified. The sample is dependent workers, identified by the EPE occupational category variable in the same way ENAHO's category variable identifies them, and the wage is monthly earnings from the primary occupation. Formality, where used, is EsSalud contributory registration. That approximates ENAHO's formality definition. Roughly 45 percent of the individuals interviewed in a given month reappear one quarter later, and the same primary sampling units return almost completely at a twelve-month lag, while there is no overlap at one, two or six months. I resample entire PSU trajectories, meaning every record a cluster contributes across every window in the estimation span, drawn once. This preserves the recurring households and individuals and the window overlap without modelling the rotation scheme. Intervals are 95 percent throughout.

[1] The increases take effect in April or May, so a policy-change year mixes interviews conducted under the old floor with interviews conducted under the new one, which mechanically raises the share recorded below the contemporaneous minimum. These records sit below the new floor and make up the zone the estimator reads, subject to the *jornada* qualification of Section 2.1.

3.3 EPEN National Quarterly Data

The national employment survey (EPEN, *Encuesta Permanente de Empleo Nacional*) publishes four disjoint calendar quarters per year, not a moving quarter released monthly, so adjacent windows share no interview months. Seventeen national quarters are available, from 2022Q1 to 2026Q1, which brackets the January 2025 increase with ten pre-policy quarters and five post-policy quarters. Because the increase took effect on the first day of a quarter, no window mixes pre-policy and post-policy months, which the Lima series cannot avoid.

The pre-policy window opens in 2022Q3, not 2022Q1. The May 2022 increase falls inside the first two quarters of the series, so the constant-floor window on S/1,025 begins only afterwards. The sample is dependent workers on the same concept ENAHO and EPE use, although this survey merges *empleados* and *obreros* into a single category code, and the wage is monthly earnings from the primary occupation. Between 22,000 and 27,000 dependent workers with positive earnings enter each quarter, drawn from roughly 6,300 primary sampling units.

Two features of the file matter. The quarterly expansion factor is missing for a share of dependent workers that grows from zero in 2022 to 8.8 percent by 2026Q1, concentrated in urban areas and in the lowest stratum, and the workers it drops earn less than those it retains. The expanded working-age population rises smoothly across the whole span with no break where the omissions begin, and the expanded counts of employed and of dependent workers behave the same way. Calibration fixes marginal totals and leaves the shape of the distribution untouched. Second, the official informality marker ships only with the annual and departmental files, so formality is not split in this survey and the primary sample is all dependent workers, as in Lima.

3.4 What Each Survey Can Date

The three series answer different timing questions. EPEN is the one that dates the 2025 increase without ambiguity, because its quarters are disjoint and the floor rose on a quarter

boundary, so no window mixes treated and untreated months.

ENAHO is an annual cross-section. Comparing a pre-policy year with a post-policy year measures a medium-run difference in the national wage distribution.

EPE is a rolling three-month series for Lima. It dates the movement to a window, because each published window pools three consecutive interview months and a window that straddles a policy date mixes treated and untreated months. What it delivers is the ordering: whether the distribution had already moved by the first fully post-policy window, or whether it continued to move afterwards.

4 Empirical Strategy

4.1 Pre-Post Distributional Estimator

I use a pre-post distributional estimator inspired by the framework of Cengiz et al. (2019), adapted for a national MW setting without control jurisdictions. Cengiz et al. (2019) construct the counterfactual using U.S. states that did not raise their MW. Peru has a single national MW. For the gross missing and excess masses and the descriptive ratio R, I use the pre-period wage distribution with a background correction from the upper tail ($> 2\times\text{MW}_{\text{new}}$) instead. The signed estimands do not use that correction. They are measured against the three counterfactual distributions defined in Section 4.2, of which the static baseline is the unadjusted pre-period distribution. The closer methodological precedent is Harasztosi and Lindner (2019), who study Hungary's national MW.

The wage distribution is discretized into S/25 bins over the range S/0–S/6,000, weighted by survey expansion factors. For each bin b, I compute the share of the workforce in the pre-period (s_b^{pre}) and post-period (s_b^{post}), and compute the raw delta: $\Delta_b = s_b^{\text{post}} - s_b^{\text{pre}}$.

The counterfactual correction uses an inverse-absolute-weighted average of Δ_b in the "clean zone" $\mathcal{C}$ (the set of bins with centers above $2 \times \text{MW}_{\text{new}}$), where the MW is unlikely to

affect the distribution:

$$\bar{\Delta}_{\text{clean}} = \frac{\sum_{b \in \mathcal{C}} w_b \cdot \Delta_b}{\sum_{b \in \mathcal{C}} w_b}, \quad w_b = \frac{1}{|\Delta_b| + \epsilon}$$

where $\epsilon = 10^{-8}$. The adjusted delta is $\Delta_b^{\text{adj}} = \Delta_b - \bar{\Delta}_{\text{clean}}$.

Bins with $|\Delta_b| \approx 0$ receive higher weight because they represent wage regions where the pre-to-post change is minimal, producing a near-zero correction that places greatest weight on bins with little observed movement. The weighting prevents a small number of bins with anomalously large movements, which may reflect sector shocks and not MW effects, from dominating the correction. Table A4 shows results are robust to uniform weighting and alternative clean zone thresholds.

Missing mass (negative-only bins in the affected zone):

$$B^{\text{miss}} = \sum_{\substack{b \in [0.85 \times \text{MW}_{\text{old}},\ \text{MW}_{\text{new}}) \\ \Delta_b^{\text{adj}} < 0}} \left| \Delta_b^{\text{adj}} \right| \tag{1}$$

Excess mass (positive-only bins in the excess zone):

$$B^{\text{exc}} = \sum_{\substack{b \in [\text{MW}_{\text{new}},\ \text{MW}_{\text{new}} + 250) \\ \Delta_b^{\text{adj}} > 0}} \Delta_b^{\text{adj}} \tag{2}$$

The paper's estimands are the **signed zone changes**. Let s_b denote the share of the reference population in wage bin b after the increase and s_b^0 the counterfactual share for the same bin. That population is formal dependent workers in the annual national data and all dependent workers in the two quarterly series, so the estimands share a definition but not a population, and the two are never pooled. Write $\mathcal{A}$ for the depletion zone, the bins with centres in $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$, $\mathcal{E}$ for the accumulation zone, the bins with centres in

$[\text{MW}_{\text{new}}, \text{MW}_{\text{new}} + \text{S}/250)$, and $\mathcal{O}$ for every remaining bin. Then

$$\Delta_A = \sum_{b \in \mathcal{A}} (s_b - s_b^0), \qquad \Delta_E = \sum_{b \in \mathcal{E}} (s_b - s_b^0), \qquad \Delta_O = \sum_{b \in \mathcal{O}} (s_b - s_b^0), \tag{3}$$

reported in percentage points. The two reported estimands are Δ_A and Δ_E. The remainder $\Delta_O = -(\Delta_A + \Delta_E)$ is an accounting residual: the three zones partition the support, and both s_b and s_b^0 sum to one, so the three signed changes sum to zero by construction and in every bootstrap replication. A binding floor implies $\Delta_A < 0$ and $\Delta_E > 0$. These are the quantities that carry confidence intervals in the distributional estimator. The department-level regressions of Section 4.4 report their own coefficients, intervals and tests, under a separate identifying assumption and with the inference described in Section 4.3.

The **bunching ratio** $R = B^{\text{exc}}/B^{\text{miss}}$ is reported alongside them as a descriptive accounting statistic. Its components take only the negative part of $s_b - s_b^0$ within $\mathcal{A}$ and only the positive part within $\mathcal{E}$, so both are non-negative by construction and will generally be positive even when the counterfactual is correct and the deviations are pure noise. R is never tested against unity or against zero.

This estimator does not require between-unit parallel trends and is computed for all four events, with Event C interpreted descriptively (Appendix F). The upper tail of the distribution ($> 2 \times \text{MW}_{\text{new}}$) enters in two places and only two. It supplies the background correction behind the gross masses and the descriptive ratio R, and it is the only region the clean-bin factor baseline is fitted on. The static pre-period and matched-calendar baselines use the observed pre-period distribution over its whole support, and the CLR-trend baseline fits bin-specific trends across every bin, not the clean zone alone, so neither takes the upper tail as its counterfactual. Where the assumption binds, it requires that movement above twice the new floor stand in for background wage growth in the MW-relevant range. The placebo tests in Section 8 check that correction: if it were biased, false-positive bunching signals would appear at artificial thresholds.

Substantial or adversely directed boundary misalignment distorts the measured zone masses, and the size of the distortion depends on where the threshold falls inside the bin. When the misalignment is severe, the bunching spike at $\mathrm{MW}_{\mathrm{new}}$ straddles a bin boundary, and records sitting exactly at the new floor are classified as "missing" instead of excess, causing severe downward bias in R. S/25 is the only displayed bin width aligned for both Event A ($\mathrm{MW}_{\mathrm{new}} = 850$) and Event C ($\mathrm{MW}_{\mathrm{new}} = 1{,}025$). Section 8 reports the sensitivity across five bin widths.

4.2 Counterfactual Baselines

Every signed change in Equation (3) is measured against a counterfactual share s_b^0. Each survey uses three baselines: the static pre-period in the annual data, or the matched calendar in the two quarterly series, together with the CLR-trend and clean-bin-factor baselines. Every table of signed zone changes reports all three for its own survey. The exception is the bin-grid sensitivity of Table 6, which holds the baseline at the matched calendar because what it varies is the grid.

Static pre-period. s_b^0 is the observed pre-period share in bin b. This is the simplest baseline and assumes the distribution would have been unchanged absent the increase. It is the ENAHO variant.

Matched calendar. The quarterly analogue of the static baseline, constructed differently in the two quarterly series because their windows differ. In the EPE series, which publishes a rolling three-month window and carries seasonality, s_b^0 is the observed share in the most recent pre-event window with the same calendar month triple, in place of the immediately preceding window. In the national quarterly series, whose calendar quarters are disjoint, the analogue is the same quarter number in the most recent pre-policy year, so that a first-quarter target is matched to the last pre-policy first quarter and not to the preceding fourth quarter.

CLR trend. Shares are mapped to centred log-ratio coordinates, where CLR denotes

the centred log-ratio transform. A bin-specific linear trend in survey time is fitted across the training periods, over every bin, not the clean zone alone, and the bin-specific slopes are shrunk toward their common mean by a ridge penalty. The fitted values are mapped back to shares by a softmax, which guarantees a valid distribution.

Clean-bin factor. A one-factor model is fitted in the same log-ratio space, but only on clean bins above twice the new floor. The factor is the leading right singular vector of the centred clean-bin log-ratio matrix, its loadings are obtained by least squares on a constant and that factor, and the loading implied by the target period's own clean bins is then used to predict the full distribution. This baseline lets the counterfactual respond to whatever moved the upper tail, without letting the affected zone inform its own counterfactual.

Three implementation choices apply to both model-based baselines and are fixed. The training observations are every available period up to and including the pre-period, so in the annual data they are event-specific: 2013 to 2015 for Event A, 2013 to 2017 for Event B and 2013 to 2021 for Event C. For Lima they are the twelve pre-event rolling windows and for the national quarterly series the ten pre-event quarters. The ridge penalty is fixed at $\lambda = 100$. The centred log-ratio transform is undefined at zero, so a constant $\epsilon = 10^{-6}$ is added to every share before taking logarithms, which matters only for empty bins in the far tails and is applied identically to training and target periods.

Reported ranges are taken from the well-centred baselines only, for the reason given in Section 8.

4.3 Inference

The three sources of variation in this paper have different sampling structures, so they require different resampling schemes. Reported intervals use the procedures below. Bias-corrected intervals are used separately as a bootstrap-centering diagnostic in Section 8.

ENAHO distributional estimates. ENAHO is a stratified multistage sample, so resampling individuals independently would understate variance by ignoring clustering within

primary sampling units. Intervals for Δ_A, Δ_E and Δ_O come from a survey-design bootstrap that resamples primary sampling units with replacement within strata, where the stratum is the department by urban-rural cell, with 1,999 replications. The counterfactual is refitted inside every replication, so uncertainty in the baseline is carried into the interval. Reported intervals are the 2.5th and 97.5th percentiles of the replicate distribution.

EPE distributional estimates. EPE publishes overlapping three-month windows, and its rotation returns the same households one quarter later and the same primary sampling units at a twelve-month lag. Resampling window by window would therefore break the dependence that the design creates. Instead I resample entire primary sampling unit trajectories, meaning every record a cluster contributes across every window in the estimation span, drawn once with replacement and rescaled by the Rao-Wu-Yue scheme, which draws $m_h = n_h - 1$ primary sampling units per stratum with replacement (Rao and Wu, 1988; Rao et al., 1992), with 999 replications. This preserves both the recurring units and the two-month overlap between adjacent windows. Because the published stratum variable is a constant for Lima, resampling is unstratified.

EPEN distributional estimates. The national quarterly series publishes disjoint calendar quarters, so adjacent windows share no interview months, but its primary sampling units recur: 55 percent return one quarter later and 71 percent return at the annual lag. Trajectory resampling is therefore required for the same reason it is in Lima, and I draw whole primary sampling unit trajectories with the same Rao-Wu-Yue rescaling and 999 replications. Here the published stratum does vary, so resampling is stratified on each unit's modal stratum. Unstratified resampling is reported as a sensitivity and widens the intervals by at most 5 percent.

Department-level regressions. With 25 departments, asymptotic cluster-robust inference is unreliable. All p-values and intervals for the regressions in Section 7.2 come from a wild cluster bootstrap-t with Rademacher weights and the null imposed, 9,999 replications, clustered on department, with intervals obtained by test inversion. Cluster-robust standard

errors are reported alongside the coefficients as a descriptive summary of dispersion and are never used to construct a test.

4.4 Share-of-Treated DiD Specification

Alongside the bunching estimator, I estimate a regression-based difference-in-differences using the department-level share of formal dependent workers in the affected wage zone as treatment intensity. Let d index departments, i index individuals, and t index calendar years. The pooled cross-sectional DiD is

$$y_{i,d,t} = \beta\, E_d \cdot \mathbb{1}[t = \text{post}] + \theta_t + X_i'\tau + \gamma_d + \varepsilon_{i,d,t}, \tag{4}$$

where $y_{i,d,t}$ is the outcome, E_d is continuous pre-policy exposure, $\mathbb{1}[t = \text{post}]$ is a post-period indicator, θ_t are year fixed effects, X_i are individual controls (age, age-squared, gender, education) with coefficient vector τ, and γ_d are department fixed effects. Exposure E_d is the department fraction of formal dependent workers in the affected wage zone $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$, measured in the pre-period only and held fixed across outcomes.

There is no exposure level term. E_d is time-invariant within department, so it is absorbed by γ_d and is not separately identified. Exposure enters continuously. Coefficients are reported rescaled to a 10 percentage point increase in exposure, since a one-unit change in E_d is outside the observed range. Inference is the wild cluster bootstrap-t described in Section 4.3.

The dynamic event-study version replaces the post indicator with a vector of year interactions $\sum_s \delta_s\, E_d \cdot \mathbb{1}[t = s]$ for s in the event window, normalizing the base year coefficient to zero. The individual controls follow those used by Jaramillo Baanante (2012), whose own design is a worker-level EPE panel, unlike the department-level exposure design used here, and the sample is restricted to urban workers aged 18 to 65. The regional exposure measure has a direct antecedent in Jiménez (2023), who applies the same share of directly affected workers to the 2016 increase. Identification checks, dynamic event-study estimates

and pooled estimates are reported in Section 7.2.

5 Main Results

5.1 National Annual Estimates: Events A, B and C

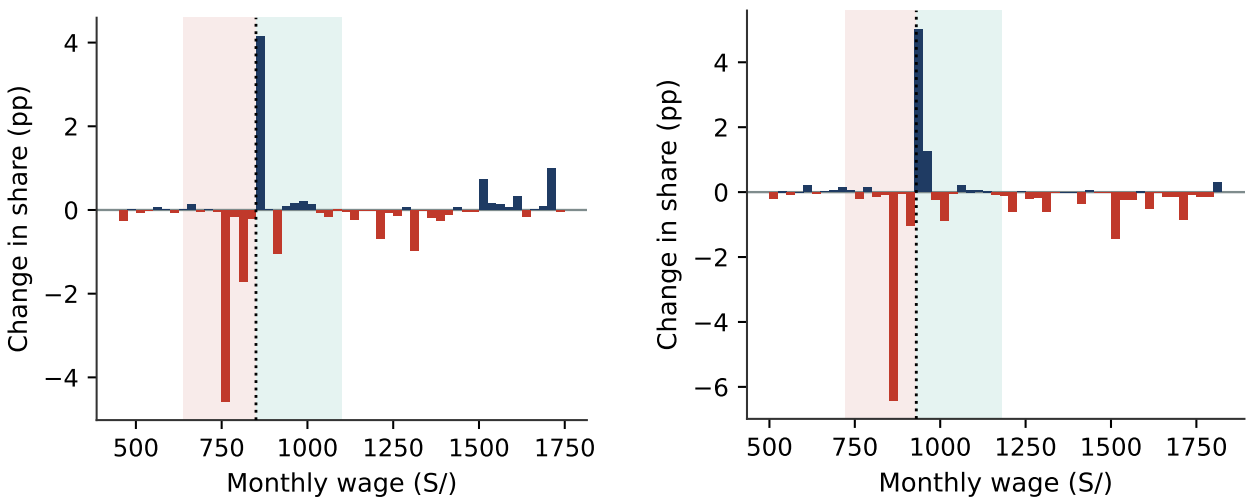
Table 3 reports the signed zone changes for all three events under each of the three counterfactual baselines. For Events A and B every Δ_E is positive and every Δ_A is negative, and no interval contains zero. Depletion below the floor exceeds accumulation above it in every cell, which means the two zones do not offset and the remainder of the distribution absorbs the difference. Event C (post-pandemic) is reported alongside them and analysed separately in Appendix F.

Table 3: Signed Zone Changes: ENAHO National Cross-Section

Event	Counterfactual baseline	Δ_E	95% CI	Δ_A	95% CI
A (750→850)	Static pre-period	+3.46	[+2.07, +4.86]	−6.54	[−7.47, −5.62]
	CLR trend	+2.32	[+1.05, +3.50]	−8.00	[−8.70, −7.39]
	Clean-bin factor	+2.12	[+0.97, +3.47]	−8.21	[−8.81, −7.33]
B (850→930)	Static pre-period	+5.42	[+3.97, +6.90]	−7.70	[−8.69, −6.72]
	CLR trend	+4.90	[+3.59, +6.13]	−9.08	[−9.81, −8.55]
	Clean-bin factor†	+4.63	[+3.38, +5.94]	−9.55	[−10.58, −8.86]
C (930→1,025)‡	Static pre-period	+7.33	[+5.48, +9.10]	−12.77	[−14.15, −11.43]
	CLR trend	+6.84	[+5.29, +8.19]	−9.24	[−10.08, −8.54]
	Clean-bin factor	+6.27	[+4.22, +7.16]	−9.54	[−11.11, −9.34]

Note: Percentage points. Δ_E is the change in the share of formal dependent workers in the accumulation zone $[\mathrm{MW}_{\mathrm{new}}, \mathrm{MW}_{\mathrm{new}} + \mathrm{S}/250)$ and Δ_A the change in the depletion zone $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$, both defined in Equation (3). Intervals are 95 percent from a stratified survey-design bootstrap resampling primary sampling units within strata, 1,999 replications. † For Event B the clean-bin factor estimate of Δ_A is materially bootstrap-centered, so its direction is informative and its magnitude is not. Bootstrap centering is discussed in Section 8. ‡ Event C is descriptive only. See Appendix F. The bunching ratio R, the gross masses and the graphical distributional evidence are reported in Appendix G as descriptive accounting statistics and are not tested.

Figure 1 plots the post-period minus pre-period change in each wage-bin share for Events A and B. The counterfactual baselines described in Section 4 are fitted to these distributions, and the signed zone changes in Table 3 are computed from the difference between the observed post-period shares and each baseline.



(a) Event A (b) Event B

Figure 1: Signed Bin-Level Changes in the Formal Wage Distribution

Note: Each bar is the post-period share of formal dependent workers in a S/25 wage bin minus the pre-period share, in percentage points, against the static pre-period baseline. Red shading marks the depletion zone and green the accumulation zone, as defined in Equation (3). The dotted line is the new minimum wage. Summing the bars within each shaded region reproduces the static pre-period row of Table 3 exactly. Raw pre-period and post-period distributions, and Event C, appear in Appendix G.

Formal dependent workers show descriptive ratios below 1 across all three events (0.71, 0.84 and 0.83 under the canonical row-level wage rule). The excess-zone count contains records that were already above the affected zone and new entrants. Gross missing mass rises across the three events, tracking the growing bite of the floor relative to the wage distribution (Table A5).

Table 4 reports the unweighted observation counts in the affected and excess zones, confirming that the bunching signal is not driven by sparse cells. The affected zone contains 837–1,171 observations pre-policy, the excess zone contains 1,591–1,669 observations post-policy.

Table 4: Effective Sample Counts in Affected and Excess Zones

Event	N_{affected} (pre) $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$	N_{excess} (post) $[\text{MW}_{\text{new}}, \text{MW}_{\text{new}} + 250)$	N_{total} (pre)
A (750→850)	837	1,669	10,059
B (850→930)	1,037	1,591	10,895
C (930→1,025)	1,171	1,595	8,946

Note: Formal dependent workers. Unweighted counts. This table and Table 2 use the same sample definition as the distributional estimator, so N_{total} for each event equals the N reported for its pre-period year. Affected zone = the wage range below the new MW that the estimator reads. Excess zone = first S/250 above the new MW (canonical excess window). $N_{\text{excess}} > N_{\text{affected}}$ because the excess-zone count includes both records in the wage range vacated below and new entrants.

5.2 The 2025 Increase at National Quarterly Frequency

Event D is the only increase this paper observes nationally and at quarterly frequency at once. It is also the cleanest to measure. The floor rose on the first day of a quarter, so no window mixes treated and untreated months. Ten quarters of constant floor precede it, against the single pre-period year the annual data allow. And between 22,000 and 27,000 dependent workers enter each quarter.

Figure 2 shows what happened bin by bin. A single large negative bar sits on the old floor and a single large positive bar on the new one. The spike migrates from the old floor to the new one.

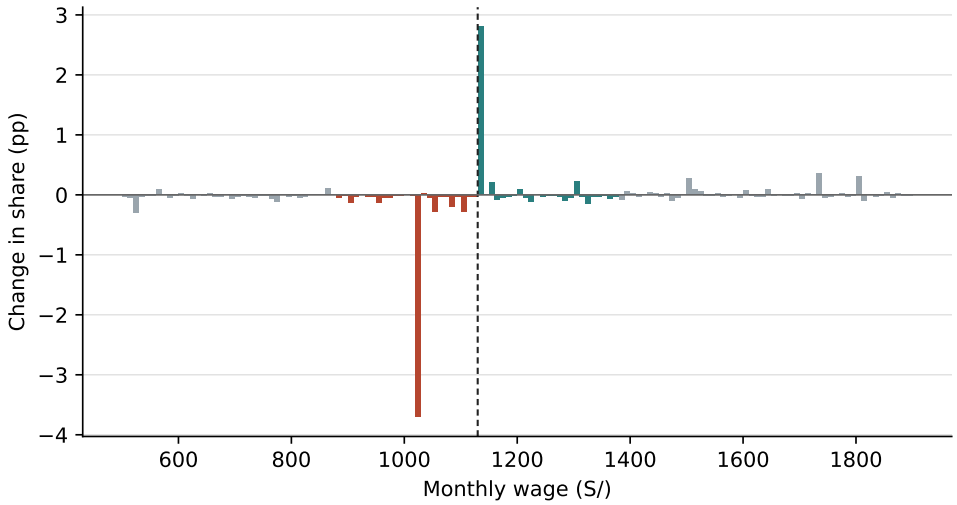


Figure 2: Signed Bin-Level Changes around the 2025 Floor

Note: Post-period mean share of dependent workers in each S/10 wage bin minus the matched-calendar counterfactual, in percentage points, averaged over the five post-policy quarters. Red marks the depletion zone and green the accumulation zone, as defined in Equation (3). The dashed line is the new minimum wage. Summing the bars within each shaded region reproduces the matched-calendar row of Table 5 exactly.

Table 5 reports the signed changes averaged over the five post-policy quarters. Across the two well-centred baselines the accumulation zone gains between 2.46 and 2.73 percentage points and the depletion zone loses between 5.04 and 5.17. The clean-bin factor baseline is materially bootstrap-centred here and is read for direction only. No interval contains zero and depletion exceeds accumulation in every cell, the same pattern as Events A and B. The intervals are among the tightest in the paper, generally around one percentage point wide.

Table 5: Event D: Signed Zone Changes, National Quarterly Series

Counterfactual baseline	Δ_E	95% CI	Δ_A	95% CI
Matched calendar	+2.46	[+1.94, +2.97]	−5.17	[−5.67, −4.69]
CLR trend	+2.73	[+2.10, +3.14]	−5.04	[−5.65, −4.69]
Clean-bin factor†	+2.06	[+1.47, +2.40]	−6.10	[−6.63, −5.89]

Note: Percentage points, averaged over the five post-policy quarters. Dependent workers, primary-occupation monthly earnings, national. S/10 bins, aligned for the new floor of S/1,130. Pre-policy window 2022Q3–2024Q4, ten quarters of constant floor at S/1,025. Intervals are 95 percent from a stratified PSU-trajectory bootstrap, 999 replications. † The clean-bin factor baseline is materially bootstrap-centred in this survey, so its direction is informative and its magnitude is not. Quoted ranges use the other two.

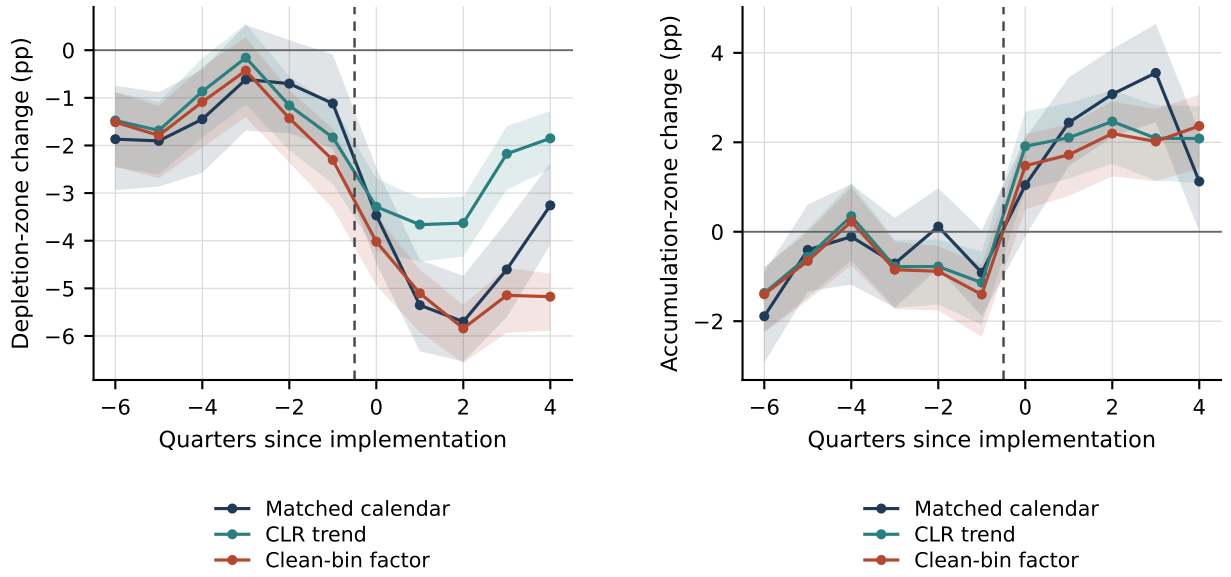
The bin grid matters here because this event tests the alignment requirement of Section 4.1 out of sample. The new floor of S/1,130 is not divisible by 25, so the paper's usual grid is misaligned for it. Table 6 reports the signed changes across four grids. The two aligned grids and the grid misaligned by S/5 agree to within 0.01 percentage points, while S/50, which misaligns by S/30, flips the sign of the accumulation estimate. Misalignment distorts in proportion to its size, and the analysis below uses the aligned S/10 grid.

Table 6: Event D: Signed Zone Changes across Bin Grids

Grid	Aligned for S/1,130?	Δ_E	Δ_A
S/5	Yes	+2.46	−5.17
S/10	Yes	+2.46	−5.17
S/25	No, by S/5	+2.47	−5.18
S/50	No, by S/30	−0.35	−2.26

Note: Percentage points, matched-calendar baseline, averaged over the five post-policy quarters, dependent workers, national. "Aligned" means 1,130 mod BinWidth = 0. At S/50 the floor falls in bin [1,100, 1,150) with centre 1,125 below 1,130, so records at the new floor fall in the depletion zone instead of the accumulation zone, which is the same mechanism Section 8 documents for Event B at S/50.

Figure 3 tracks both zones against event time under a rolling-origin fit, so that every plotted quarter is out of sample. Pre-policy accumulation estimates are non-positive, and several placebo intervals in Table A23 exclude zero on the negative side, although none produces significant positive accumulation. Both zones then move together at the first post-policy quarter and keep moving for two to three quarters more before levelling. That is a gradual profile, matching Event A in the Lima series of Section 5.3, and here there are no contaminated windows.



(a) Depletion zone

(b) Accumulation zone

Figure 3: Event D: Depletion and Accumulation by Quarter

Note: Signed zone change in percentage points against quarters since implementation, dependent workers, national. The counterfactual for each target quarter is fitted only on quarters strictly before it inside the constant-floor window, so every point is out of sample. Bands are 95 percent stratified PSU-trajectory bootstrap intervals. The dashed line marks the boundary between the last pre-policy quarter and the first post-policy one.

Two falsification exercises apply here as they do in Lima. Five pseudo-events placed inside the constant-floor window on S/1,025 produce no significant accumulation in any of the fifteen combinations of date and baseline. Moving the threshold away from the statutory value while holding the date fixed makes the statutory floor the largest positive estimate among the seven thresholds under all three baselines. One cell rejects. A threshold set 50 percent above the statutory floor is significant under the clean-bin factor at about a fifth of the statutory effect. Appendix N reports both grids in full.

In this survey the clean-bin factor is materially bootstrap-centred, by roughly half a replicate standard deviation, while the CLR trend that misbehaved in Lima is well centred here. The direction survives both interval constructions in every case, and the quoted range rests on the matched-calendar and CLR-trend baselines.

5.3 Timing and Signed Changes in the Lima Series

Table 7 reports the signed zone changes. For Event A the accumulation zone gains 3.44 percentage points under the matched-calendar baseline and 2.95 under the clean-bin factor baseline. For Event B the corresponding figures are 3.98 and 4.41. For Events A and B every interval in these comparisons excludes zero, and the depletion zone falls significantly in every case. The CLR-trend baseline yields larger estimates for both events, 5.50 and 5.34, but it is materially bootstrap-centered, so its sign carries information and its size does not. Across the two well-centered baselines the excess-zone increase ranges from 2.9 to 4.4 percentage points.

Table 7: EPE Lima: Signed Zone Changes by Counterfactual Baseline

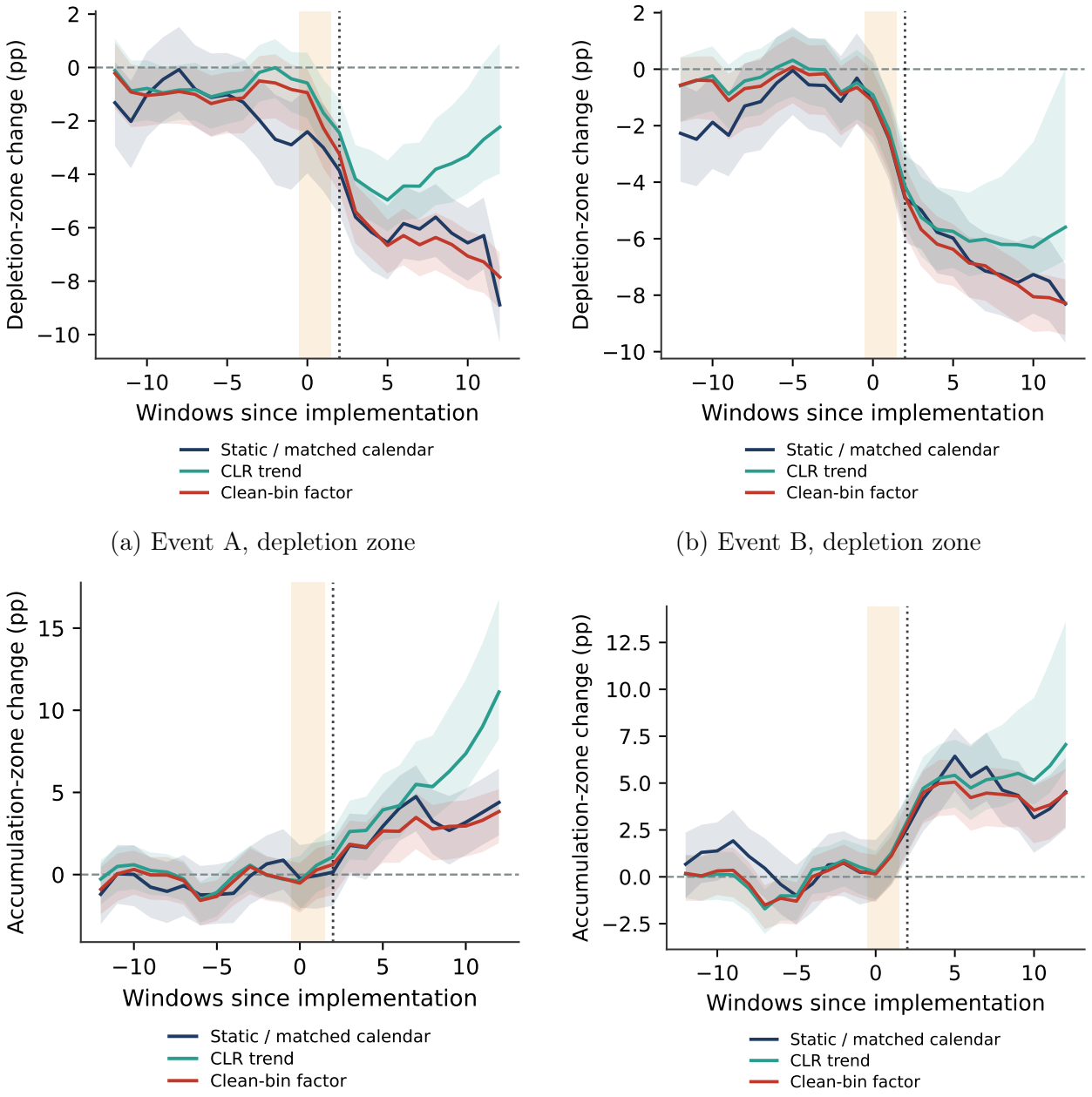
Event	Baseline	Δ_E	95% CI	Δ_A	95% CI
A	Matched calendar	$+3.44$	$[+1.59, +5.30]$	-4.76	$[-6.17, -3.35]$
	Clean-bin factor	$+2.95$	$[+1.47, +4.31]$	-5.43	$[-6.43, -4.36]$
	CLR trend†	$+5.50$	$[+3.82, +8.44]$	-3.04	$[-4.27, -1.16]$
B	Matched calendar	$+3.98$	$[+2.37, +5.32]$	-5.44	$[-6.78, -4.12]$
	Clean-bin factor	$+4.41$	$[+3.23, +5.42]$	-6.43	$[-7.44, -5.40]$
	CLR trend†	$+5.34$	$[+4.15, +8.20]$	-5.43	$[-6.46, -2.92]$
C	Matched calendar	$+0.74$	$[-1.45, +3.07]$	-5.54	$[-7.82, -3.40]$
	Clean-bin factor	$+1.06$	$[-0.82, +2.93]$	-4.61	$[-6.25, -2.87]$
	CLR trend†	$+1.50$	$[-0.55, +3.76]$	-4.39	$[-6.03, -2.15]$

Note: Percentage points, post minus pre. Dependent workers, primary-occupation monthly earnings, Lima Metropolitan Area. Unit of observation is the rolling three-month window. Intervals are 95 percent from an unstratified PSU-trajectory bootstrap, 999 replications. † The CLR-trend baseline is materially bootstrap-centered, so its sign is informative and its magnitude is not. Any quoted range should use the two well-centered baselines.

On the ordering that Section 3.4 isolates, the two events differ. Event B adjusts quickly.

The accumulation-zone change is flat through the pre-period, becomes distinguishable from zero at the first fully post-policy window, and stays at a similar level for the following year, with all three counterfactual baselines tracking one another. Event A adjusts gradually. Depletion below the floor is visible earlier and more clearly than accumulation above it, and the accumulation estimate keeps growing for roughly a further quarter before levelling off. Consistent with this, Event A's estimated accumulation is larger when measured over a longer post-policy horizon than over a short one, whereas Event B's is stable across horizons.

Figure 4 shows that contrast window by window, for both zones and all three baselines. The shaded band marks the two windows that straddle the increase and therefore mix pre-policy and post-policy months. Adjacent estimates are mechanically correlated because their windows share two interview months, which the trajectory bootstrap accounts for.



(a) Event A, depletion zone

(b) Event B, depletion zone

(c) Event A, accumulation zone

(d) Event B, accumulation zone

Figure 4: EPE Lima: Depletion and Accumulation by Rolling-Window Event Time

Note: Change in the share of dependent workers in each zone, percentage points, against rolling-window event time. Shaded band marks the two windows containing one and two post-policy interview months. Dotted vertical line marks the first fully post-policy window. Bands are 95 percent PSU-trajectory bootstrap intervals. Adjacent estimates are mechanically correlated because consecutive windows share two of their three interview months.

Two falsification exercises support the specificity of these results. Re-running the estimator at eight pseudo-event windows where no minimum wage change occurs, and excluding the disrupted fieldwork period of 2020, yields no significant accumulation in any of the eight, for either event, under any of the three baselines. Moving the threshold away from the statutory value while holding the event date fixed makes the statutory threshold the largest positive estimate among the seven thresholds, for both events under every baseline. For Event B a threshold set 15 percent above the statutory floor, at S/1,070, is also significant under two of the three baselines at roughly a third to a half of the statutory effect. That is compatible with wage spillover above the minimum.

5.4 The Four Events Side by Side

Figure 5 places all four events and all three surveys side by side. Events A and B agree across the two surveys that observe them. Event C does not (Appendix F). Event D sits at the low end of the accumulation estimates with much the tightest intervals.

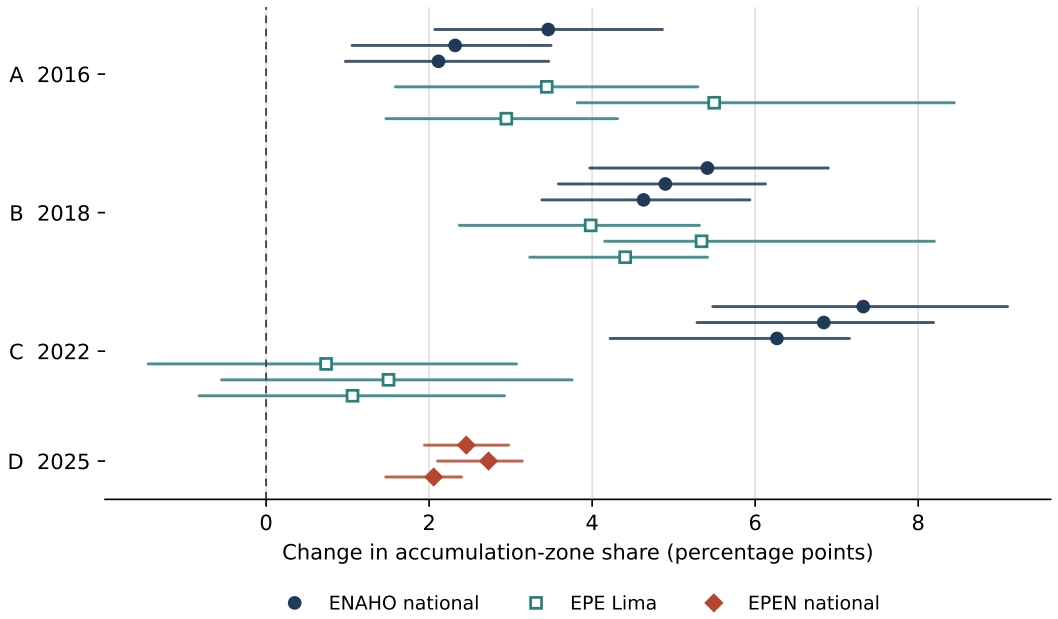


Figure 5: Accumulation-Zone Change across Events and Surveys

Note: Signed accumulation-zone change by event, survey and counterfactual baseline, with 95 percent intervals. Filled circles are annual national estimates on formal dependent workers, open squares are Lima rolling-quarter estimates on all dependent workers, diamonds are national quarterly estimates on all dependent workers. Event D is observed in one survey only, because the national quarterly series carries no pre-policy window for the 2016 and 2018 increases and a single clean pre-policy quarter for 2022.

Reading the three surveys together brings out one pattern. Excluding the baselines the table notes withhold from quoted magnitudes, and excluding the descriptive Event C, the depletion estimates on formal dependent workers run from -6.54 to -9.08 percentage points across Events A and B, and those on all dependent workers, whether national or Lima, run from -4.76 to -6.43 across Events A, B and D. Point estimates are more negative in the formal-only sample than in the all-dependent-worker samples, and the surveys and policy episodes differ.

6 Relation to the Peruvian Literature

Several earlier contributions establish the empirical and theoretical context for the estimates above. Yamada (1996) separates Peru's uncovered sector into informal self-employment and informal salaried work, finds the lowest wages in the salaried component, and reads self-employment as an option, not a refuge. That is the distinction the composition accounting in Section 7.1 relies on. Jaramillo Baanante (2004) finds short-run earnings effects confined to the bottom of the informal distribution and, in the formal sector, to workers caught between the old and the new minima, with little support for a lighthouse effect in the short run. That formal-sector finding is the closest antecedent to the affected zone this paper measures. Jaramillo and López (2006) reach a similar conclusion for the following decade, reporting a wage effect that is very limited by regional standards and confined to formal workers earning around the floor.

A second group estimates employment responses directly. Céspedes (2006) estimates an employment-RMV elasticity of about -0.13 on a sector panel running from 1997 to 2003, significant only at roughly 84 percent, and states that the estimate supports its sign and not its magnitude. He also reports that the formal wage distribution in Lima changes perceptibly in the neighbourhood of the floor, which is the pattern this paper measures directly. Céspedes and Sánchez (2014) track seven minimum wage changes between 2002 and 2011 on the EPE Lima series, finding a statistically significant minimum wage-income elasticity and employment effects that are moderate in large firms and larger in small ones. Jaramillo Baanante (2012), covering the 2000s in Lima, also finds no numeraire or lighthouse effect. His wage effects are statistically weak and small, he finds no significant effect on the probability of staying employed for workers earning around the floor, and he finds a positive employment effect for informal wage earners at least 20 percent above it, which he reads as substitution toward slightly more skilled workers, not as a contraction. His graphical analysis of that decade suggests only a limited role for the floor in the wage distribution, which the present estimates for 2016 and 2018 do not reproduce.

Two studies speak to reallocation across the formal-informal boundary, the margin this paper's composition evidence concerns. Del Valle (2009), at the Banco Central de Reserva del Perú (BCRP), applies difference-in-differences to pre-2009 adjustments and finds no aggregate effect. Disaggregating by wage range, she finds a negative effect on formal employment that is counteracted but not offset by a positive effect on informal employment, implying a partial reallocation of workers near the floor from the formal to the informal sector. That is the channel this paper's composition evidence speaks to. Corcuera García (2024) studies a Peruvian increase with firm- and worker-level data and finds that some workers are reallocated towards self-employment, the same direction as the Event B share-of-treated estimate reported here. He also reports negative employment elasticities larger than recent estimates in the literature, driven by firms competing with the informal sector, together with pass-through of the labor-cost increase to consumer prices at medium and large firms. The department-level estimates reported below show no systematic employment pattern and are too wide to adjudicate between them.

Three recent contributions bear on the present analysis from different angles. Castellares et al. (2022) estimate the effect of RMV increases on the price level using disaggregated consumer price data together with sector-specific measures of minimum wage exposure, and find that more exposed industries raise their prices by more, with the response larger where imported input content and price rigidity are lower. On their estimates a 10 percent minimum wage increase raises the consumer price index by up to 0.73 percentage points after twelve months. Price adjustment is one margin along which firms can absorb a labor-cost shock, although Corcuera García (2024) finds it coexisting with negative employment elasticities instead of substituting for them. I do not test for pass-through on the events studied here. Armas et al. (2023) document that Peru's informal labor market sets wages flexibly while the formal market is rigid, and that the informal sector's shock-absorbing role was overwhelmed by the pandemic contraction. That contrast bears directly on why an estimator restricted to formal dependent workers can miss adjustment running through the informal

margin. Jiménez (2023) studies the 2016 increase with a difference-in-differences design on the regional share of directly affected workers, the same exposure measure used here, but with approval of government performance as the outcome in place of employment.

A complementary theoretical mechanism is offered by Felix (2021), who shows that in concentrated labor markets MW increases can raise wages without reducing employment due to upward-sloping firm-level labor supply. The monopsony channel offers one account under which a binding floor can reshape the wage distribution without a large aggregate employment response.

7 Composition and Scope

7.1 Within-Zone Composition Accounting

The headline composition fact in the affected zone $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ is that the cross-sectional count of dependent workers in the zone declines while the small pool of informal self-employed workers earning in this range remains roughly constant. The surveys compare different cross-sectional samples, so what is observed is the net change in weighted counts, not the movement of individual workers. Absolute counts of dependent workers in the zone fall by 49 to 55 percent across the three events. Absolute counts of informal self-employed workers in the same wage range are roughly stable (Event A +9.1%, Event B +8.6%, Event C +0.1%). The total count of zone occupants falls by 29 to 35 percent. Table 8 documents these counts.

Table 8: Composition of the Affected Zone, Pre vs. Post: Informal Self-Employment Share Rises through Subtraction of Dependent Workers

Event & period	Inf. SE share (%)	Inf. SE count (000s)	Dep. count (000s)	Total (000s)	All-SE share (%)
Event A pre (2015)	31.35	395	808	1,260	35.85
Event A post (2017)	47.86	431	401	900	55.43
Δ (pp / 000s)	*+16.51*	*+36*	*-407*	*-360*	*+19.58*
Event B pre (2017)	29.24	403	910	1,379	34.00
Event B post (2019)	48.52	438	412	903	54.37
Δ (pp / 000s)	*+19.28*	*+35*	*-498*	*-476*	*+20.37*
Event C pre (2021)	27.33	476	1,197	1,742	31.25
Event C post (2023)	41.54	476	609	1,147	46.93
Δ (pp / 000s)	*+14.21*	*+0*	*-588*	*-595*	*+15.68*

Notes: Cells report the weighted composition of workers reporting income in the affected zone $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$. Income source: *p524a1* (monthly cash wage) for dependent workers ($p507 \in \{3, 4, 6\}$) and *i530a/12* (annual self-employment income, monthlyized) for self-employed ($p507 = 2$). Informal SE = *p507*=2 & *ocupinf*=1 (cuenta propia informal). Dep. count includes both formal and informal dependent workers. Counts in thousands, weighted by survey expansion factor (*fac500a*). All-SE share (rightmost column) is the broader definition including formal SE. The paper's main claim references the Informal-SE share (column 1), which is the margin the composition accounting concerns. These are repeated cross-sections, so the cells record net changes in weighted counts, not movements of individual workers.

Reading: In each event, the Informal-SE share rises by 14–19 pp post-policy. The Informal-SE count, however, is roughly constant (Event A: +9.1%, Event B: +8.6%, Event C: +0.1%) while the Dependent count collapses (-49% to -55%) and the Total count of zone occupants falls (-29% to -35%). The share rise is therefore driven primarily by subtraction of dependent workers from the zone, not by net entry into informal self-employment.

As a mechanical consequence, the informal-SE *share* within the affected zone rises sub-

stantially after each MW increase, from 31.4% to 47.9% in Event A (+16.5pp), 29.2% to 48.5% in Event B (+19.3pp), and 27.3% to 41.5% in Event C (+14.2pp). This share increase is primarily a denominator effect. The dependent-worker count in the affected wage range falls, the informal-SE pool earning in that range stays roughly constant, and a smaller denominator with a stable numerator produces a larger share.

A separate piece of evidence bears on composition. The department-level regressions in Section 7.2 find that for Event B more exposed departments saw larger increases in informal self-employment. That pattern does not appear for the other two increases, and it is sensitive to the exclusion of Lima.

The decline in the affected-zone dependent-worker count is consistent with three non-exclusive margins of adjustment: (i) movement above the new MW (the bunching margin documented above), (ii) transition to non-employment, and (iii) transition into other informal employment arrangements, including informal self-employment, informal dependent employment, or family work. Cross-sectional ENAHO data cannot decompose the decline into these specific destinations because workers are not tracked across years. The ENAHO panel of Appendix A follows treated and control workers from 2021 to 2023 and reports employment retention, formal dependent retention, wage growth and hours for the survivors. It does not decompose the decline into destinations. The within-zone compositional pattern is evidence of a decline in the affected-zone dependent-worker count.

7.2 Employment Composition: the Event B Exposure Gradient

The share-of-treated difference-in-differences specified in Equation (4) provides a regression-based test for the aggregate composition shift that the within-zone accounting of Table 8 records as a decline in dependent-worker counts. That accounting is in weighted counts, whereas the distributional estimator of Equation (3) is in shares, and the regression works on a different sample at a different level of aggregation again. The identification rests on parallel trends between high and low share-of-treated departments. I assess this with two

complementary placebo specifications on the 2013–2015 window, in which the minimum wage remained constant at S/750 and no other major shock occurred. The standard placebo uses Event A's actual share-of-treated dose (computed from the 2015 affected zone) and tests aggregate outcomes on 2013–2015. The apples-to-apples placebo redefines the dose using the 2013 wage-range-specific zone $[0.85 \times 750, 750)$ and tests the same two aggregate outcomes together with informal self-employment, which the Event A dose does not test. Neither placebo rejects on the outcomes it covers, as Panel A of Table A18 reports, so the only placebo evidence on the informal self-employment outcome comes from the wage-range dose. With 25 departments these joint tests are a diagnostic. The Kaitz instrumental-variable specification also yields first-stage F-statistics below the Stock-Yogo threshold in every event (Table A3), reflecting the structurally limited cross-sectional variation in the Kaitz index across Peru's 25 departments under a national wage floor. I therefore report the share-of-treated reduced form in place of IV estimates. Panel-based identification using the ENAHO rotating panel (Module 978) is constrained by 76 percent two-year attrition. Aggregate attrition rates are similar across treatment and control groups (Appendix A). I report the panel retention and wage statistics as a supplementary analysis in Appendix A.

For Events A and B, the employment and formal-employment event studies show no common post-policy sign or persistent exposure gradient. Joint lead restrictions do not reject for either event on either outcome (Table A18, Panel B), but the confidence intervals remain wide. These estimates rule out a precisely estimated common employment response. Both event studies, and the Event C panels, appear in Appendix C.

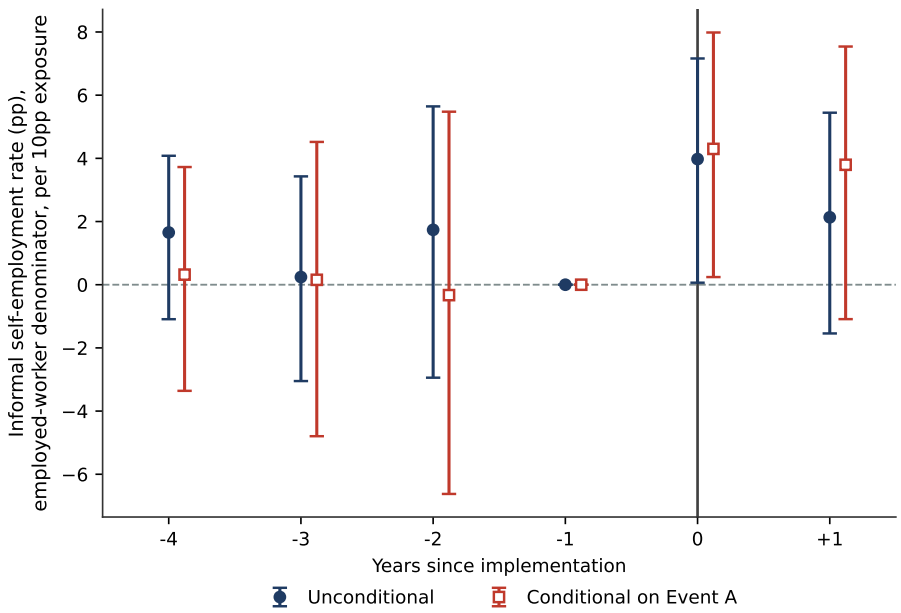


Figure 6: Informal Self-Employment Response to Event B, per 10 Percentage Points of Exposure

Note: Dynamic event-study coefficients on continuous pre-policy exposure, informal self-employment rate with an employed-worker denominator, base year 2017, department fixed effects, no exposure level term. Coefficients scaled to a 10 percentage point increase in exposure. Intervals are 95 percent wild cluster bootstrap-t by test inversion, Rademacher weights, null imposed, 9,999 replications, clustered on department (25 clusters). Individual controls are age, age squared, gender and education. The sample is urban workers aged 18 to 65 with positive survey weights. The conditional series adds Event A exposure interacted with year dummies. Events A and C appear in Appendix C.

Figure 6 documents a positive informal self-employment gradient following Event B. A 10-percentage-point increase in exposure is associated with a 3.98 percentage-point increase in 2018 (95 percent interval $[0.06, 7.16]$). Conditioning on exposure to Event A yields a similar estimate of 4.30 points ($[0.24, 7.98]$). The gradient remains positive in 2019, although the interval includes zero, and the joint lead restrictions do not reject under either specification (Table A18, Panel B). Its attenuation when Lima is excluded (Table A18, Panel C) and its absence in Events A and C indicate an episode-specific adjustment. Event A's joint lead restriction rejects under the same continuous specification ($p = 0.032$), and no statistically

distinguishable informal-self-employment gradient is detected for Event C.

Lima is the minimum-endpoint department in every leave-one-out specification, so a substantial share of the gradient is carried by one department.

The reallocation towards self-employment that Corcuera García (2024) finds with firm- and worker-level data runs in the same direction as the Event B gradient, which is the one worker-level outcome in Section 6 directly comparable to an outcome estimated here. On employment that paper reports negative elasticities larger than recent estimates in the literature, and the department-level intervals here are too wide to speak to them.

The distributional estimator and the department-level regressions measure different objects, on different samples and at different levels of aggregation, so they are read as two independent pieces of evidence.

The same estimator applied to informal dependent workers does not isolate a policy response, because its behaviour there tracks the local shape of the informal wage distribution. Appendix L reports that exercise and what it implies for the lighthouse hypothesis of Jaramillo Baanante (2004).

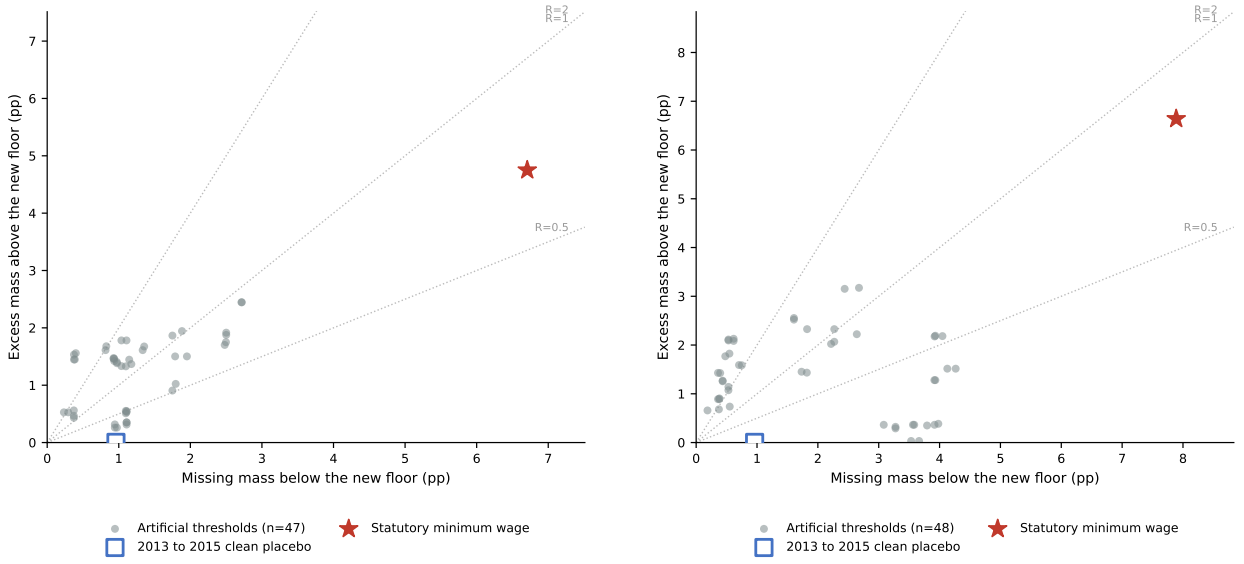
8 Robustness and Validation

Each counterfactual baseline is diagnosed for bootstrap centering in each survey, and the affected one is reported in full and excluded from every quoted magnitude. The diagnosis differs across surveys. In ENAHO the clean-bin factor baseline is materially bootstrap-centered in Δ_A for Event B, by roughly a quarter of a replicate standard deviation. Its direction is stable under both percentile and bias-corrected intervals. The static pre-period and CLR-trend baselines are clean to modestly centered across events. In EPE the problem sits elsewhere. There the CLR-trend baseline is materially centered in Δ_E, by 0.6 to 1.0 percentage points, because it extrapolates ridge-shrunk bin-specific trends in log-ratio space and maps them back through a softmax. Its sign survives both interval constructions but

its magnitudes do not, and it produces the largest EPE point estimates. In the national quarterly series the affected baseline is the clean-bin factor again, centred by roughly half a replicate standard deviation. Each survey is assessed on its own, and in every case the affected baseline is reported in full and no headline magnitude is quoted from it.

I test the bunching estimator's identification along two complementary dimensions. The first dimension tests whether the bunching signal could appear at arbitrary wage points absent policy. Applying the same estimator to artificial thresholds on the Event B population, using fake increases from S/1,100→S/1,200 and S/1,400→S/1,500, produces ratios far below the statutory value, with missing mass four to six times smaller than at the true MW. Panel A of Table A6 reports these exercises. The bunching signal is specific to the actual MW threshold and does not appear at arbitrary other wage points.

The second dimension tests whether the estimator could detect a bunching signal in a window without policy at the true MW threshold. I apply Event A's threshold structure ($\text{MW}_{\text{old}} = 750$, $\text{MW}_{\text{new}} = 850$) to the 2013–2014 ENAHO cross-section, a single-year window in which the nominal MW was constant at S/750 throughout. Panel B of Table A6 reports the comparison. The placebo recovers almost no excess mass at S/850, against a large gain at the same threshold in Event A, so no spike forms at the future floor in a window where the floor did not move. Missing mass in the placebo is well below Event A's but is far from zero. Some mass leaves the zone through ordinary wage growth, so the depletion side of the estimator is not a clean policy quantity in the way the accumulation side is. This asymmetry is why the paper reads Δ_A and Δ_E together and never as a ratio, and why the counterfactual baselines in Section 4 are fitted to trends instead of to a single pre-period level.



(a) Event A

(b) Event B

Figure 7: Gross Missing and Excess Mass across Aligned Thresholds

Note: Each grey point is one artificial threshold pair aligned to the S/25 bin grid, with the estimator run exactly as at the statutory value. The red star is the statutory minimum wage. The open square is the 2013 to 2015 window, in which the nominal floor did not change. That window matches the two-year horizon of the events themselves, so it is a different exercise from the single-year 2013 to 2014 placebo reported in the text. Axes are the gross masses defined in Section 4.1, which are non-negative by construction and are not tested. The statutory threshold is extreme on both dimensions simultaneously, which no artificial threshold achieves. Event C's raw wage distribution and descriptive ratio statistics appear in Appendix G.

Table A4 presents counterfactual sensitivity for all three events under six specifications. Canonical results are stable: alternative clean zone thresholds (1.5× and 2.5×) and no background correction all yield identical R across events, indicating that the inverse-absolute-weighted background correction is near zero. Uniform weighting lowers R by 0.025 to 0.064 across events. The degree-4 polynomial exceeds one for Events B and C, and the estimator becomes less reliable as MW bite grows. Across the five non-polynomial specifications, the range of R is 0.644–0.708 (A), 0.780–0.842 (B), and 0.808–0.833 (C).

The canonical inverse-absolute-weighted background correction is close to zero across all

events, vanishing at fourth-decimal precision in all three (Table A5). This near-zero finding is in part a property of the weighting scheme. Inverse-absolute weighting down-weights bins with large $|\Delta_b|$ and up-weights bins with small $|\Delta_b|$, so the weighted mean of clean-zone deltas is mechanically pulled toward zero whenever the clean zone contains a mix of small and large deviations. Clean-zone drift is better measured by the unweighted (uniform) mean of the clean-zone deltas, reported in Table A5. Each mean is computed across approximately 160 to 170 clean bins above $2 \times \text{MW}_{\text{new}}$, and each is positive, reflecting a modest rightward shift of the upper tail of the formal wage distribution between pre- and post-periods consistent with economy-wide wage growth. Applied as a uniform per-bin correction, the ratio falls from 0.708, 0.842 and 0.833 under the inverse-absolute correction to 0.644, 0.780 and 0.808 under uniform weighting. The ratio remains in the same qualitative range under either weighting scheme, and the canonical near-zero correction is partly a property of the weighting scheme.

A degree-4 polynomial is fitted to the post-period wage distribution in the clean zone ($> 2 \times \text{MW}_{\text{new}}$) and extrapolated into the affected zone as the counterfactual level. Because the upper-tail polynomial is fitted outside the affected zone, it can extrapolate a high counterfactual into the missing zone, making the observed post-distribution appear to lie *above* the counterfactual there. This generates artificially small missing mass (the zone that "should" have workers according to the polynomial actually does) and consequently inflates R above 1. This is a limitation of polynomial counterfactuals in bunching settings with sharp mass points (Kleven and Waseem, 2013): the polynomial cannot distinguish between the smooth background trend and the mass concentration created by the MW itself. The inverse-absolute procedure excludes the MW region from its clean zone and down-weights large deviations within that upper-tail zone, producing a counterfactual that reflects only smooth background drift.

The discrete-bin construction of the missing and excess zones imposes the sensitivity anticipated in Section 4.1. At S/50 bins, for example, $\text{MW}_{\text{new}} = 930$ falls in bin $[900, 950)$ with center $925 < 930$, and workers earning S/930–S/949 are counted in the *missing* zone instead

of the excess zone, collapsing R from 0.842 to 0.206, a 76% downward bias indistinguishable from real employment destruction. Table 9 reports the bunching ratio across five bin widths and indicates whether each is boundary-aligned for each event.

Table 9: Bin Width Sensitivity: Bunching Ratios (Formal Dependent Workers)

Bin width	Boundary aligned? A / B / C	R (Event A)	R (Event B)	R (Event C)
S/10	Y / Y / N	0.756	0.841	0.847
S/15	N / Y / N	0.091[†]	0.855	0.848
S/20	N / N / N	0.723	0.820	0.853
S/25 (baseline grid)	Y / N / Y	0.708	0.842	0.833
S/50	Y / N / N	0.691	0.206[†]	0.661

Note: Canonical inv-abs background correction, excess window S/250, formal dependent workers. "Boundary aligned" = MW_{new} mod BinWidth = 0, so the MW threshold falls on a bin edge. [†] Boundary misalignment: the bunching spike at MW_{new} falls in a bin whose center lies below MW_{new}, classifying it as missing instead of excess. See text.

The bin-boundary alignment requirement (Section 4.1) is confirmed in Table 9: among the displayed aligned widths, Event A varies from 0.691 to 0.756 and Event B from 0.841 to 0.855, while Event C has only one aligned displayed width, S/25, so an aligned-width range is not available for it. Among aligned widths R varies by 0.014 for Event B and by 0.065 for Event A. Severely misaligned widths produce large distortions ($R = 0.091$ for Event A at S/15, and $R = 0.206$ for Event B at S/50). Event B at S/25 is marginally misaligned by S/5, which does not detectably distort R.

Event C's pre-period falls under post-pandemic conditions. Aggregate formal employment fell 9 percent in 2020 and had recovered only partially by 2021. I use 2021 as the pre-period because it is eight months before the May 2022 increase and because the nominal floor of S/930 applies in both 2021 and 2022, so the comparison does not straddle a change in the statutory value. The formal dependent wage distribution in the affected zone

[791, 1,025) is close to identical in 2019 and 2021, at 520,000 and 511,000 workers with a median of S/930 in both years. The pandemic did not visibly compress the part of the distribution this estimator reads.

Event C is not identified through the department-level design. Its event-study coefficients reject the joint pre-trend restriction for both employment and formal employment, consistent with differential post-pandemic recovery, and it is analysed separately in Appendix F.

9 Conclusion

Peru raised its minimum wage four times between 2016 and 2025, from S/750 to S/1,130, a cumulative 51 percent in nominal terms. Each increase left a band of workers earning below a floor that had just moved above them. What became of that band is the question here, and two quantities answer it: how much of the share earning below the new floor emptied out, and how much of it reappeared in the band immediately above. A binding floor implies that the first falls and the second rises.

For the 2016, 2018 and 2025 increases the distribution does what a binding floor implies. The share below the new minimum falls and the share just above it rises, under every counterfactual baseline and in every survey that observes the episode, and those surveys differ in sample, frequency, geographic scope and formality definition. In the annual national data the gain just above the floor runs from 2.1 to 5.4 percentage points across baselines and events, in Lima from 2.9 to 4.4, and in the national quarterly series for 2025 from 2.5 to 2.7.

The two do not offset. In every event, under every baseline and in every survey, the band below the floor loses more than the band just above it gains. The 2025 increase measures that gap best, being the only one observed nationally and at quarterly frequency at once, with ten quarters of constant floor behind it and an effective date on the first day of a quarter, so that no observation window mixes months under the old floor with months under the new one. There the band above the floor gains 2.5 to 2.7 percentage points while the

band below it loses 5.0 to 5.2, and the bin-level picture is a single large decline on the old floor and a single large gain on the new one. The rest of the wage distribution accounts for the difference between the two bands. The department-level evidence locates part of it. After the 2018 increase, more exposed departments saw larger increases in informal self-employment, a gradient specific to that episode and carried substantially by Lima, which points to a formal-to-informal margin as one route out of the band.

The high-frequency series also date the movement, which annual data cannot. The 2018 adjustment is already present in the first fully post-policy window and stays there, while the 2016 adjustment keeps building for roughly a further quarter and the 2025 adjustment for two to three quarters more. Adjustment to a new floor is therefore not instantaneous.

The 2022 increase, which falls in the post-pandemic recovery, behaves differently from the other three and is treated as descriptive throughout.

The design is a before-and-after comparison, since Peru's floor leaves no untreated region.

Administrative wage registries at firm-month frequency would follow individual workers across the threshold and locate the rest of the share that leaves the band. The national quarterly design used here for 2025 will be available for every increase from here on, with a longer pre-period each time.

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A ENAHO Panel 978: Event C

The ENAHO panel 978 tracks a subsample of individuals from 2021 to 2023. Treatment: formal dependent workers in 2021 with wages in [S/791, S/1,025). Control: wages in [S/1,230, S/2,563]. Approximately 76% of both groups are not re-interviewed. Results apply to the surviving 24% only. Outcome statistics are weighted by the panel expansion factor. Attrited workers are *not* counted as non-employed.

Table A1: Panel Transition Results (Event C, 2021→2023)

Metric	Treatment ($N = 1{,}171$)	Control ($N = 3{,}903$)	DiD
Re-interviewed rate	24.4%	23.6%	—
Among re-interviewed (weighted):			
Employment retention	63.9%	60.1%	+3.8pp
Formal dep retention	46.8%	49.9%	−3.1pp
Δ log wage (stayers)	+0.307	+0.154	+0.153
Δ hours (stayers)	+0.7h	−1.3h	+2.0h

Note: 76% attrition over two years. Attrition rates are balanced across treatment and control, at 75.58 and 76.40 percent, a difference of 0.82 percentage points against a sampling error of approximately 1.4 percentage points. The chi-square test of independence across the full transition matrix yields $p = 0.022$, reflecting differences in destination cells, not overall attrition selectivity.

B Non-Identified Employment Designs

Two employment designs were attempted and neither is identified. This appendix records them.

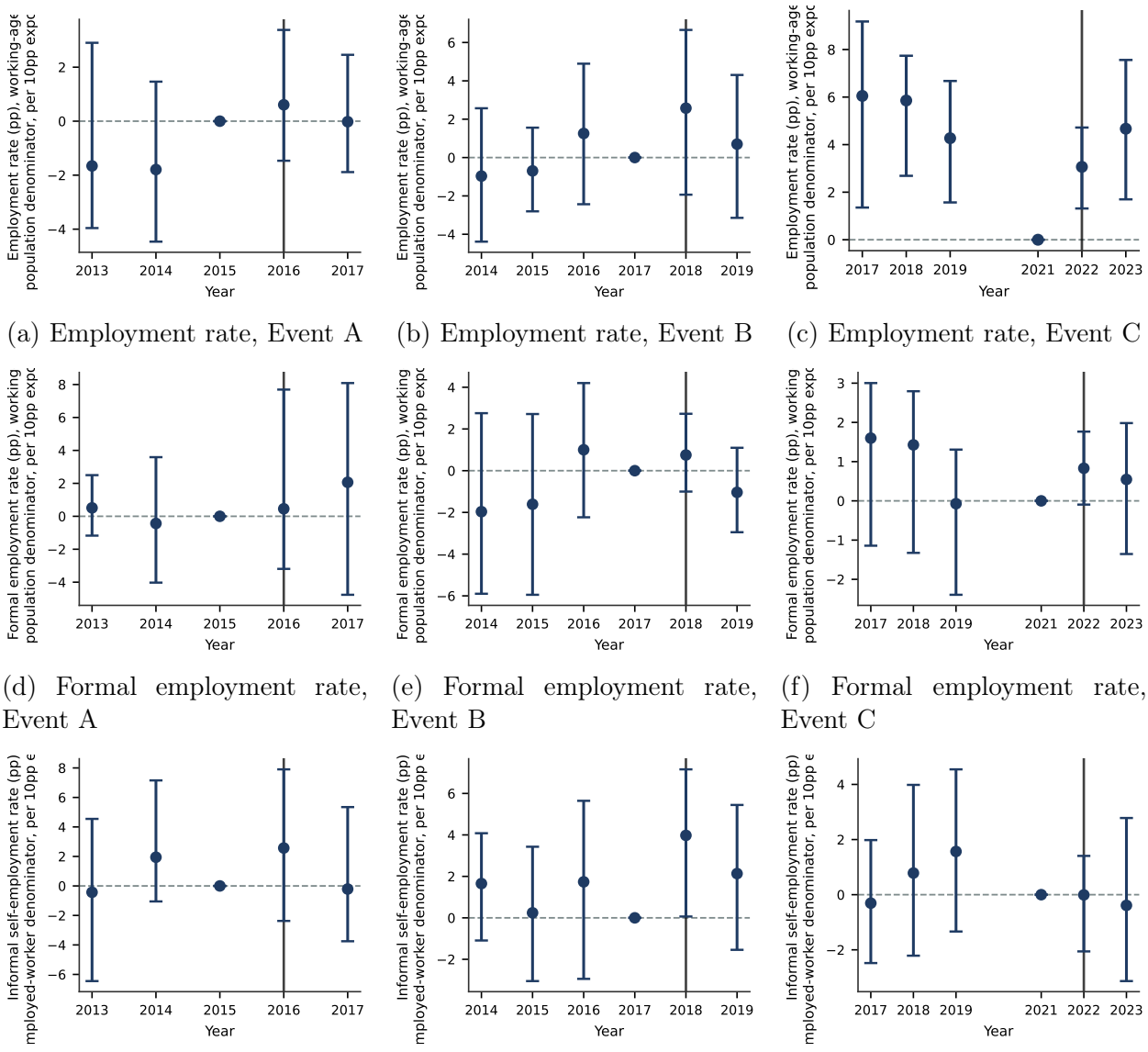
The first is the department-level share-of-treated event study using windows that span a preceding policy change. Event B rejects the joint pre-trend restriction under such windows, but the windows overlap the adjustment to the 2016 increase, so the rejection cannot be

attributed to either event. Narrowing the windows to avoid the overlap removes the rejection but also removes most of the identifying variation. The narrow-window specifications are the ones displayed in Appendix C and tabulated in Appendix N.

The second is the Kaitz instrument, reported in Appendix D. The department-level version is weak in every event against the conventional thresholds of Stock and Yogo (2005). Separately, the within-year province version reaches a strong first stage, but a placebo year with no policy change produces a comparably strong one, which indicates that specification is tracking seasonal wage dynamics, not minimum-wage exposure. Table A3 reports all three constructions.

C Department-Level Event Studies and the Event C Confound

The employment and formal-employment event studies for Events A and B appear here instead of in the main text, since neither identifies a response and the main text needs only the summary given in Section 7.2.



(a) Employment rate, Event A (b) Employment rate, Event B (c) Employment rate, Event C

(d) Formal employment rate, Event A (e) Formal employment rate, Event B (f) Formal employment rate, Event C

(g) Informal self-employment rate, Event A (h) Informal self-employment rate, Event B (i) Informal self-employment rate, Event C

Figure A1: Department-Level Event Studies by Outcome and Event

Note: Continuous pre-policy exposure interacted with year dummies, base year normalized to zero, department fixed effects, no exposure level term. Coefficients are scaled to a 10 percentage point increase in exposure. Intervals are 95 percent wild cluster bootstrap-t by test inversion, Rademacher weights, null imposed, 9,999 replications, clustered on department (25 clusters). Employment and formal employment use a working-age population denominator, and informal self-employment uses an employed-worker denominator. Individual controls are age, age squared, gender and education. The sample is urban workers aged 18 to 65 with positive survey weights. Panel (h) shows the unconditional Event B specification. The version conditioning on Event A exposure is in Figure 6.

Event C's event-study coefficients are kept out of the main text because the joint lead restriction rejects under the canonical wild cluster bootstrap for both employment and formal employment (Table A18, Panel B), consistent with differential post-pandemic recovery by department and not a minimum-wage response. The post-treatment coefficients therefore conflate the May 2022 minimum wage increase with macroeconomic recovery from the 2020 employment collapse. The three outcome rows in Figure A1 display the full A–B–C event-study sequence using the same specification and palette as Figure 6 in Section 7.2.

For employment, the pre-period coefficient on 2019 is already +4.27 percentage points per 10 percentage points of exposure ($p = 0.001$), and post-treatment coefficients for 2022 and 2023 are +3.06 and +4.67 respectively. High share-of-treated departments diverged from low share-of-treated departments by approximately the same magnitude before the May 2022 policy change as after it, consistent with the recovery being the underlying driver. For formal employment, the post-treatment coefficients are +0.83 and +0.55 percentage points, neither distinguishable from zero, and the joint lead restriction rejects for that outcome as well. For self-employment, the coefficient is approximately flat and turns slightly negative in 2023 (-0.39 percentage points per 10 percentage points of exposure, $p = 0.740$), consistent with the post-pandemic recovery pulling some workers from informal self-employment back into formal dependent employment.

The formal employment rate on a working-age population denominator shows the same pattern for Event C. Leads reject, so the post-policy coefficients cannot be read as a policy response.

Informal self-employment for Events A and C is shown next. Neither replicates the Event B gradient reported in the main text.

D Department-Level Kaitz IV: Structural Weakness from National-MW Setting

Three IV approaches were tested, and Table A3 reports the first stage of each. The department-level Kaitz instrument falls far below the conventional Stock-Yogo threshold of 10 in all three events. The province-level version clears that threshold for Event B but carries a reversed first-stage sign and fails pre-trends. The within-year province version has the strongest first stage of the three, but a placebo year with no policy change produces a comparably strong one, so that specification captures seasonal wage dynamics in place of minimum-wage exposure.

Table A2: Department-Level Kaitz IV

Event	$\hat{\pi}$ (FS)	SE	F	$\hat{\beta}$ (RF)	SE	OWE	SE
A	+0.169	0.082	4.29	+0.014	0.038	+0.082	0.229
B	−0.070	0.103	0.46	+0.122	0.044	—	—
C	+0.132	0.090	2.16	+0.220	0.076	+1.667	1.271

Note: FS = first stage, RF = reduced form, OWE = own-wage elasticity. Event B OWE not reported (reversed first-stage sign). SE clustered by department (25 clusters). All F fall far below the conventional Stock-Yogo threshold of 10. Instrument is weak.

Table A3 sets the three constructions side by side on first-stage strength alone. Only the within-year province design produces a strong first stage, and the placebo row directly below it shows that the same design produces one in a year with no policy change. That is the reason none of the three is used for identification.

Table A3: Instrument Strength across Three IV Constructions

Construction	Unit	Event	First-stage F
Kaitz, department level	Department	A	4.29
	Department	B	0.46
	Department	C	2.16
Kaitz, province level	Province	B	11.72
Kaitz, within-year province	Province	B	16.15
placebo, no-change year	Province	—	12.71

Note: Conventional Stock-Yogo rule-of-thumb threshold is 10. The province-level construction clears the threshold but carries a reversed first-stage sign and fails pre-trends. The within-year province construction clears it in both the policy year and a year with no policy change, indicating that it tracks seasonal wage dynamics, not minimum-wage exposure. Full coefficients for the department-level construction appear in Table A2. No specification in this table is used for identification in the paper.

D.1 Cross-Department Kaitz Variation

The Kaitz index (MW divided by the weighted-median formal wage by department) ranges from 0.39 to 1.09 across departments and years, with substantial change in the spatial ranking between events. In 2015 (Event A pre-period), Ica had the highest Kaitz (0.94), driven by formal agricultural workers on agro-export farms earning near-minimum monthly wages despite full-time hours (median 48 hours/week), while Huancavelica had a low Kaitz (≈ 0.50) because approximately 85% of its formal workers were public-sector employees earning above the MW. By 2021 (Event C pre-period), the ranking had shifted substantially. Appendix P maps the 2021 distribution and discusses the inter-event evolution.

This variation reflects differences in economic structure, mainly the shares of agriculture and public-sector employment. Differential MW exposure does not drive it, and it does not generate sufficient first-stage power for IV identification (the first stages fall far below

10, see Table A3). Within Kaitz-based event-study specifications, employment pre-trends appear violated for Events B and C when using windows that span the rolling adjustment of preceding policy changes (Appendix B and Section 7.2 document that placebo windows outside any rolling adjustment do not reject the lead restrictions). The Kaitz index is used in this paper as a descriptive measure of MW bindingness, not as an instrument.

E EPEN Ciudades Annual: Single-Period Bunching at the Floor

This exercise is separate from everything else in the paper and answers a different question: does the floor produce a visible spike in a single cross-section? No pre-post comparison is available in this survey, because EPEN carries no pre-policy window for the 2016 and 2018 increases and only four pre-policy months for 2022. Using the EPEN Ciudades annual file for 2023 (INEI code 873), restricted to formal dependent employed workers (N = 53,316), a single-period bunching estimator in the tradition of Lee (1999) and Saez (2010), with a degree-four polynomial counterfactual fitted outside a S/75 exclusion window around the floor yields an excess-mass factor of 1.34 at the minimum wage, with 1.14 percent of formal dependent weight sitting in the MW bin itself. Every city in the file shows excess mass at the floor.

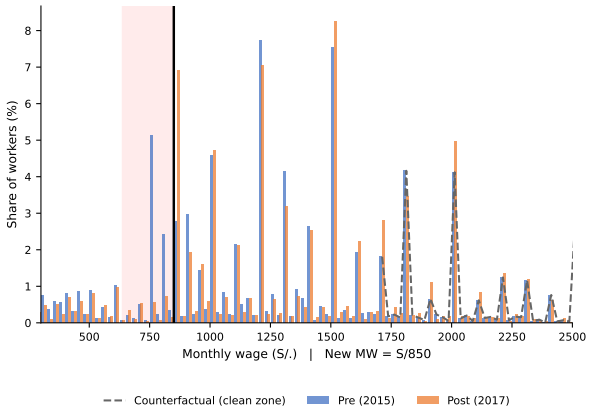
It carries no confidence interval, so it is descriptive only. It is a single-period estimate, so it speaks to whether the floor binds. The 2023 file is post-Event C and is not evidence on Event C, whose distributional movement is treated in Appendix F. Cross-sectional excess mass at the floor is independent corroboration, from a different survey and a different estimator family, that the minimum wage binds in Peru's formal wage distribution.

F Event C Limitations

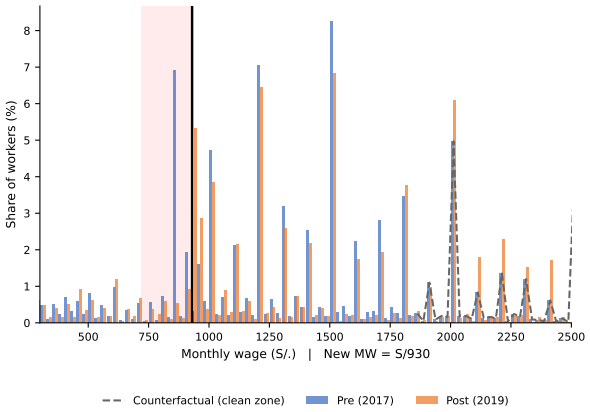
Event C is not comparable to the two earlier increases and is reported as descriptive evidence throughout. Three facts establish this. Its department-level event studies fail the joint pre-trend restriction sharply for both employment and formal employment, consistent with differential post-pandemic recovery, not a wage-floor response. In the Lima high-frequency data the accumulation-zone change is not distinguishable from zero under any of the three baselines or either interval construction, while the depletion-zone change is negative and significant, but the Lima data detect no robust pile-up above the floor. And the Lima estimation window is thin: EPE was redesigned from August 2022, which leaves exactly one clean fully post-policy rolling window, and the pre-period placebo behaves poorly, returning 7, 3 and 5 positive rejections across the twelve pre-event windows under the three baselines against roughly 0.6 expected at the 5 percent level, driven by large unexplained swings in late 2021.

The national annual and Lima high-frequency evidence also disagree on this event. ENAHO gives an accumulation-zone increase of 6.3 to 7.3 percentage points depending on baseline, while the Lima estimates are 0.7 to 1.5 with intervals spanning zero. A geographic story, a timing story and a formality-definition story are all available, and none is established by these data. Event C's national annual distributional movement is descriptive, and the high-frequency Lima evidence does not detect robust accumulation above the new floor.

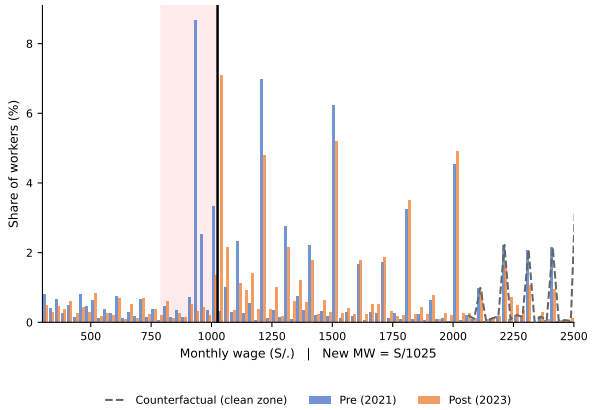
G Descriptive Ratio Statistics



(a) Event A: S/750 to S/850 (May 2016)



(b) Event B: S/850 to S/930 (April 2018)



(c) Event C: S/930 to S/1,025 (May 2022)

Figure A2: Wage Distributions Before and After MW Increases

Note: Each panel shows the formal dependent wage distribution (S/25 bins, weighted) before (blue) and after (orange) the MW increase, with panel-specific pre and post years in each legend. The vertical solid black line marks the new MW. The new MW value is also shown in each panel's x-axis label. Shaded region (pink) is the affected zone, defined as $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$. Dashed grey line in the clean zone above $2 \times \text{MW}_{\text{new}}$ is the canonical counterfactual (pre-period shares plus the inverse-absolute-weighted background drift, near zero in all events). Y-axes show workforce share (percent).

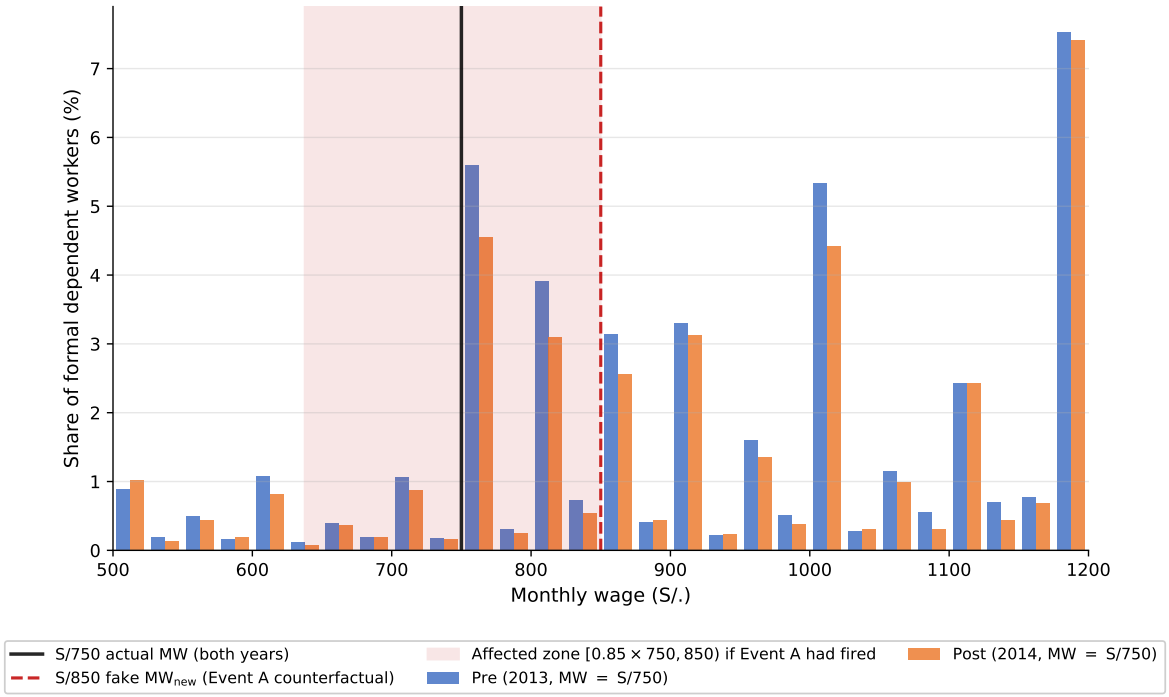


Figure A3: Placebo at the true threshold: formal dependent wage distribution, 2013 vs 2014

Note: Weighted wage histogram of formal dependent workers in 2013 (blue) and 2014 (orange). MW was constant at S/750 throughout the window. The S/750 spike (solid black line) is present in both years, as expected. No spike emerges at S/850 (dashed red line), the counterfactual fake MW_{new} from Event A's threshold structure. The natural rightward drift of the distribution accounts for the placebo's +2.38pp baseline missing mass but does not produce a spurious bunching spike at the counterfactual threshold.

The bunching ratio and the gross zone masses are reported here as descriptive accounting statistics. They are not tested and support no inferential or causal claim in the main text. Both gross masses take only the signed part of the deviation within their zone, so both are non-negative by construction and generally positive under a correct counterfactual with noisy deviations.

Table A4: Counterfactual Sensitivity: Bunching Ratios, All Events

Specification	Clean zone / Weighting	R (Event A)	R (Event B)	R (Event C)
Canonical (inv-abs)	$> 2 \times$ MW, inv-abs	0.708	0.842	0.833
Uniform weights	$> 2 \times$ MW, uniform	0.644	0.780	0.808
Polynomial deg. 4	$> 2 \times$ MW, poly fit	0.456	1.056	1.889[†]
Clean at $1.5 \times$ MW	$> 1.5 \times$ MW, inv-abs	0.708	0.842	0.833
Clean at $2.5 \times$ MW	$> 2.5 \times$ MW, inv-abs	0.708	0.842	0.833
No correction	none	0.708	0.842	0.833

Note: Formal dependent workers. Missing zone: $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$ (negative-only bins). Excess window: $[\text{MW}_{\text{new}}, \text{MW}_{\text{new}} + 250)$. Background correction varies by specification. Canonical and no-correction specifications give identical R because the clean-zone background is near zero for all three events. [†] Polynomial overcorrection: the degree-4 fit extrapolates through the peak of the distribution, inflating the counterfactual in the missing zone. Event C's $R = 1.889$ reflects severe overcorrection for the post-pandemic Event C increase (10.2%). See text.

G.1 Diagnostic and Falsification Statistics

Table A5 collects the diagnostic quantities that Section 5 refers to. Both gross masses grow across the three events, tracking the rising bite of the floor, and the displayed ratio is the displayed excess divided by the displayed missing in each column. The drift row reports the unweighted mean of the clean-zone deltas, next to the canonical inverse-absolute correction, which is near zero partly as a property of the weighting scheme.

Table A5: ENAHO Bunching Diagnostics by Event

Diagnostic	Event A	Event B	Event C
Gross missing mass at statutory MW (pp)	6.71	7.89	12.92
Gross excess mass at statutory MW (pp)	4.75	6.64	10.76
Descriptive ratio R	0.708	0.842	0.833
Clean-zone drift, uniform mean (pp per bin)	+0.042	+0.045	+0.022
Inverse-absolute background correction (pp)	+0.0000	+0.0000	+0.0000

Note: Formal dependent workers, S/25 bins, canonical row-level wage rule. Gross missing mass is the sum of negative adjusted bin deltas over the affected zone $[0.85 \times \mathrm{MW}_{\mathrm{old}}, \mathrm{MW}_{\mathrm{new}})$ and gross excess mass the sum of positive adjusted deltas over $[\mathrm{MW}_{\mathrm{new}}, \mathrm{MW}_{\mathrm{new}} + 250)$. Both are non-negative by construction, so neither is tested against zero, and R is their ratio. Clean-zone statistics are computed across approximately 160 to 170 bins above $2 \times \mathrm{MW}_{\mathrm{new}}$.

Table A6 reports the two falsification exercises. Panel A moves the threshold pair away from the statutory floor while holding the population fixed, and Panel B holds the threshold structure fixed while moving to a window in which the floor did not change. The excess side collapses in the placebo window while the missing side does not, which is why the paper reads the two signed changes together and never as a ratio.

Table A6: Falsification: Artificial Thresholds and a Constant-Floor Window

Exercise	Threshold pair	Missing (pp)	Excess (pp)	R
Panel A: artificial thresholds, Event B population				
Statutory floor	S/850→S/930	7.89	6.64	0.842
Artificial	S/1,100→S/1,200	1.33	0.08	0.057
Artificial	S/1,400→S/1,500	1.98	0.03	0.016
Panel B: constant-floor window, Event A threshold structure				
Event A, 2015–2017	S/750→S/850	6.71	4.75	0.708
Placebo, 2013–2014	S/750→S/850	2.38	0.06	0.026

Note: Panel A applies the estimator to the 2017–2019 population at threshold pairs where no minimum wage change occurred. Panel B applies Event A's threshold structure to the 2013–2014 cross-section, in which the nominal floor was constant at S/750. Missing and excess masses are gross quantities and are never tested against zero. Each displayed R is the displayed excess divided by the displayed missing.

H Hours (Descriptive)

The intensive-margin hours response is measured by a 2×2 DiD of weighted mean weekly hours. The treatment group is workers in the affected wage zone $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$ in the pre-period and $[\text{MW}_{\text{new}}, 1.3 \times \text{MW}_{\text{new}}]$ in the post-period (covering the wage range in which post-policy bunching may appear). The control group is workers in $[1.5 \times \text{MW}_{\text{new}}, 3.0 \times \text{MW}_{\text{new}}]$ in both periods.

Table A7 presents the intensive-margin hours DiD.

Table A7: Weekly Hours DiD

Event	Treat pre	Treat post	Δ treat	Ctrl pre	Ctrl post	Δ ctrl	DiD
A (750→850)	44.4h	44.8h	+0.4h	42.1h	41.5h	−0.7h	+1.1h
B (850→930)	44.8h	45.1h	+0.3h	41.2h	42.0h	+0.8h	−0.5h
C (930→1,025)	44.5h	46.4h	+1.9h	42.1h	43.6h	+1.5h	+0.4h

Note: Treatment pre = workers in $[0.85 \times \text{MW}_{\text{old}}, \text{MW}_{\text{new}})$. Treatment post = workers in $[\text{MW}_{\text{new}}, 1.3 \times \text{MW}_{\text{new}}]$ (bunched-up zone). Control = $[1.5 \times \text{MW}_{\text{new}}, 3.0 \times \text{MW}_{\text{new}}]$. Weighted means using survey expansion factors. Outcome is total weekly hours worked.

No DiD exceeds ±1.1h/week. The descriptive hours differences are small, ranging from −0.5 to +1.1 hours per week.

Median weekly hours among workers at the new floor are close to those of workers above it. This is a comparison of different individuals before and after the increase, reported as a descriptive fact about the hours distribution.

I Residual Compression under Stated Assumptions

The decomposition below rests on assumptions about arrival wages and counterfactual wage growth, and the component labelled residual is description.

Table A8 presents the wage compression DiD with a decomposition into mechanical (composition) and residual components.

Table A8: Wage Compression DiD with Composition Decomposition

Event	Total DiD	Mechanical	Residual	% Mech.	New arrivals
A (750→850)	−0.035	−0.028	−0.008	78%	13% of zone
B (850→930)	−0.050	−0.021	−0.029	41%	10% of zone
C (930→1,025)	−0.069	−0.064	−0.005	92%	26% of zone

Note: Compression zone = $[\text{MW}_{\text{new}}, 1.5\times\text{MW}_{\text{new}}]$. High-wage zone = $[2\times\text{MW}_{\text{new}}, 3\times\text{MW}_{\text{new}}]$. Mechanical = predicted DiD from new arrivals entering at $\log(\text{MW}_{\text{new}})$ while existing workers grow at the high-wage rate. Residual = total minus mechanical. With individual controls (age, sex, education, sector), the compression coefficient is −0.045 ($p < 0.001$). Derived columns are computed from unrounded inputs, so they may differ in the last digit from the difference of the displayed rounded values.

For Event B, approximately 41% of the −0.050 DiD reflects the composition effect, that is the mass the accounting exercise classifies as entering the compression zone and diluting its mean log wage. The remaining 59% (−0.029 log points) represents residual relative wage compression. For Events A and C, the composition effect dominates (78% and 92%). The compression DiD is a descriptive statistic of distributional change. The distributional patterns I document are consistent with Engbom and Moser (2022), who find that Brazil's national MW compresses the wage distribution from below, and with Jales (2018), who uses a density discontinuity approach to estimate MW effects in Brazil.

Compression robustness. The canonical mechanical decomposition assumes new arrivals enter at exactly $\log(\text{MW}_{\text{new}})$ and existing compression-zone workers grow at the same rate as the high-wage control group. Table A9 relaxes both assumptions for Event B. Moving the assumed entry point from log(930) to log(980) or log(1,030) *reduces* the mechanical share (from 41% to 31% and 21%), increasing the residual compression. This is because higher entry wages are less dilutive: new arrivals pull the mean down less, so more of the observed compression must reflect residual suppression. Similarly, if existing workers grew at only

50% of the high-wage rate, the mechanical share rises marginally to 46%. In all four cases, residual compression remains negative and economically meaningful (-0.027 to -0.040 log points), confirming that the compression result is not an artifact of the canonical entry-point assumption.

New arrivals as a share of the post-period compression zone vary substantially across events: 13% in Event A, 10% in Event B, and 26% in Event C (Table A8). Event C's higher share of new arrivals (26%) reflects both the S/95 increase, which was larger than Event B's S/80, and the post-COVID recovery in formal employment: the formal workforce grew approximately 11% between 2021 and 2023, bringing a larger cohort of workers into formalization near the new floor of S/1,025. The high mechanical share in Event C (92%) is therefore an artifact of this post-pandemic compositional shift. The mechanical decomposition assumes that new entrants' wages are determined independently of the MW, so that they enter at the floor because it is the market wage in their sector, not because the MW creates a mass point. If the MW increase drove formalization of previously informal workers (formalization channel), these entrants' wages are set by the MW itself, and classifying their compositional contribution as "mechanical" understates residual compression. The 92% mechanical share is therefore the canonical scenario estimate of the non-MW-related component, and the 8% is the residual under that same scenario. Neither is a bound.

Table A9: Compression Robustness: Mechanical vs. Residual (Event B)

Assumption	Mechanical	Residual	% Mech.
1. Canonical: new arrivals at log(930)	−0.021	−0.029	41%
2. New arrivals at log(980)	−0.015	−0.035	31%
3. New arrivals at log(1,030)	−0.010	−0.040	21%
4. Existing workers grow at 50% of control rate	−0.023	−0.027	46%

Note: Event B (S/850→S/930, pre=2017, post=2019). Formal dependent workers. Total DiD fixed at −0.050 (observed). New arrivals = 10% of compression zone post-period ($n_{\text{new}}/n_{\text{post}} = 0.10$). Mechanical = predicted DiD contribution from composition change. Residual = total − mechanical. The high-wage growth rate $H = +0.005$ log points (from data). Entry points in assumptions 2–3 represent workers entering above the exact MW floor. Derived columns are computed from unrounded inputs, so they may differ in the last digit from the difference of the displayed rounded values.

J Subgroup Ratio Comparisons (Descriptive)

The subgroup comparisons below are computed on the bunching ratio, and their sampling variation across events is large relative to any difference between subgroups.

The organizing question is whether subgroups with higher MW exposure show systematically different bunching ratios. Table A10 presents selected comparisons. Full tables by industry, firm size, age/sex, and geography appear in Appendix K.

Table A10: Bunching Ratios by Subgroup (Selected Comparisons)

Group	Event A	Event B	Event C	Average
Geography				
Lima Metropolitan	0.643	1.064	0.815	0.840
Rest of country	0.847	0.620	0.889	0.785
Employer type				
Public sector	0.924	0.455	0.891	0.756
Private sector	0.775	0.974	0.764	0.838
Firm size				
Micro (≤ 10)	0.894	0.909	0.694	0.833
Medium/Large (51+)	0.693	0.799	0.883	0.792
Industry				
Manufacturing	1.001	0.933	1.054	0.996
Education/health	0.600	0.723	0.949	0.757

Note: Geography, employer-type and firm-size ratios use the canonical row-level wage rule, the same sample as the distributional estimator, and the all-groups series under that rule reproduces 0.71 / 0.84 / 0.83. All eight groups use that rule, with industry classified from the CIIU rev4 code in p506r4. Table 3 reports signed zone changes, not ratios. Full subgroup tables in Appendix K.

The evidence does not support a consistent enforcement or exposure gradient across events. The private sector has a higher average descriptive ratio across these three episodes than the public sector (0.838 vs. 0.756), which could reflect market adjustment more than inspection-driven compliance. Lima's ratio takes values 0.643 (Event A), 1.064 (Event B), and 0.815 (Event C), a range of 0.42 across stable geographic boundaries within a six-year window. This within-Lima variability is large relative to the across-events range of the national estimate (0.708 to 0.842) and points to subsample-specific dynamics that the aggregate estimator averages over. Plausible drivers include the Venezuelan migration shock

peaking around 2018, construction-sector cycles tied to specific infrastructure projects, and Lima-specific informal absorption capacity that fluctuates with macro conditions. The data do not disentangle these. Micro firms show higher ratios than large firms in Events A–B, the *opposite* of an enforcement prediction. One reading is self-selection, in that micro firms that formalize may already have internalized compliance costs. Agriculture, mining and utilities has the highest average ratio (0.997) but on a range from 0.479 to 1.879, manufacturing averages marginally below it (0.996) and is the only sector to fall between 0.93 and 1.06 in all three events, and education and health has the lowest average (0.757), which one might read as public-sector wage-scale rigidity that constrains adjustment. Several subgroups differ sharply from the aggregate in individual events, agriculture, mining and utilities reaching 1.879 in Event A against a national 0.708, but no difference is stable across all three events.[2]

[2]I also examined heterogeneity by ethnicity. Among formal dependent workers, indigenous (Quechua/Aymara) and non-indigenous workers have similar wage distributions (both median S/1,400) and MW exposure (8.5% vs. 10.7% below MW). Bunching ratios by ethnicity are unstable across events (indigenous average 1.27, range 0.63–1.80) and reflect sector composition (59% public sector) and not differential MW compliance. The ethnic divide operates through access to formal employment, with an indigenous formality rate of 5.7% vs. 20.7% for Spanish speakers (see Chávez, 2024).

K Heterogeneity: Full Tables

Table A11: Bunching Ratios by Industry Sector

Sector	Event A	Event B	Event C	Average
Total ($N = 10{,}059$)	**0.708**	**0.842**	**0.833**	**0.794**
Agriculture/Mining/Utilities ($N = 1{,}430$)	1.879	0.479	0.634	0.997
Manufacturing ($N = 921$)	1.001	0.933	1.054	0.996
Commerce ($N = 951$)	0.772	1.202	0.719	0.898
Transport/hospitality ($N = 563$)	1.037	1.054	0.571	0.887
Finance/professional services ($N = 997$)	0.724	0.933	0.737	0.798
Public administration ($N = 1{,}652$)	1.027	0.507	0.834	0.789
Education and health ($N = 3{,}045$)	0.600	0.723	0.949	0.757

Note: $N = 2015$ pre-period sample under the canonical row-level wage rule, the same sample as Table 3. Industry is classified from the CIIU rev4 code (*Clasificación Industrial Internacional Uniforme*, revision 4) in `p506r4`, taken at the 2-digit division. Workers with a missing code fall outside the named sectors but remain in the total. Construction excluded (sparse cells, < 500 obs). Agriculture/Mining includes CIIU rev4 codes 35–39 (utilities).

Subgroup comparisons by firm size, worker age and sex follow.

Table A12: Bunching Ratios by Firm Size, Age, and Sex

Group	Event A	Event B	Event C	Average
Panel A: Firm size				
Micro (≤ 10, $N = 710$)	0.894	0.909	0.694	0.833
Small (11–50, $N = 1{,}250$)	0.910	0.916	0.600	0.809
Medium/Large (51+, $N = 7{,}631$)	0.693	0.799	0.883	0.792
Panel B: Age and sex				
Age 18–24	0.685	1.014	0.792	0.830
Age 25–44	0.754	0.793	0.803	0.783
Age 45–64	0.739	0.966	0.861	0.856
Men	0.745	0.798	0.776	0.773
Women	0.688	0.934	0.884	0.835

Note: Firm size measured by the reported establishment worker count. Workers with missing firm-size data (4.3–5.8% per year) are excluded from Panel A. The three rows sum to 95.3% of the sample.

The final subgroup cut is by geography and employer sector.

Table A13: Bunching Ratios by Geography and Employer Sector

Group	Event A	Event B	Event C	Average
Lima Metropolitan ($N = 2{,}637$)	0.643	1.064	0.815	0.840
Rest of country ($N = 7{,}422$)	0.847	0.620	0.889	0.785
Urban ($N = 9{,}079$)	0.717	0.854	0.832	0.801
Rural ($N = 980$)	0.771	0.735	0.822	0.776
Public sector ($N = 4{,}350$)	0.924	0.455	0.891	0.756
Private sector ($N = 5{,}647$)	0.775	0.974	0.764	0.838

Note: Public sector identified by employer type (central, regional, or local government) or CIIU 2-digit codes 84–86 (public administration, education, health).

L The Informal Sample and the Lighthouse Hypothesis

Informal dependent workers show ratios of 1.22, 0.99 and 0.81 across the three events. These are descriptive quantities. What evidence there is on informal-sector reallocation comes from the department-level regressions in Section 7.2, and it is confined to Event B.

Jaramillo Baanante (2004) argues that informal-sector wages cluster at the formal MW through a "lighthouse" effect, in which even workers outside the formal MW's legal scope adopt it as a focal reference point. I applied the bunching estimator to the informal sample at four sham threshold pairs covering the wage distribution: S/600→S/700, S/750→S/850, S/1,100→S/1,200, and S/1,400→S/1,500, all measured from the 2018 to the 2019 cross-section, a pair of years carrying the same statutory floor of S/930 (Table 2) and containing no change in the statutory value between them. The resulting R values span $[0.77, 9.36]$, with no consistent pattern.

The actual informal R values for Events A, B, and C (1.22, 0.99, 0.81) fall entirely within this sham range. The high-range sham ($R = 0.77$, threshold S/1,400→S/1,500) sits at a higher segment of the informal wage distribution comparable to Event C's threshold range ($R = 0.81$), and the low-range sham ($R = 9.36$, threshold S/600→S/700) sits at a lower segment comparable to Event A's range ($R = 1.22$). The pattern in the real Events' informal R values appears to track the local geometry of the informal wage distribution and not a consistent policy response. This indicates that the estimator cannot distinguish a policy response from that geometry.

I interpret the dispersion across sham thresholds as identification failure in the informal sample. The estimator's behavior depends sharply on which segment of the wage distribution the threshold pair covers. Heaping at round-number spikes, density gradients across the informal wage distribution, and other features unrelated to policy generate spurious bunching signal that varies by hundreds of percent across plausible sham thresholds. The lighthouse hypothesis is neither confirmed nor refuted by the cross-sectional design, and both Jaramillo Baanante (2004) and Jaramillo Baanante (2012) report finding no such effect on

Peruvian data with designs better suited to testing it.

The lighthouse hypothesis concerns the response of informal *wages* to the formal MW threshold and is distinct from the response of informal *employment composition* to the policy. The regression-based evidence in Section 7.2 documents, for Event B only, a positive composition shift toward informal self-employment in more exposed departments, which speaks to worker reallocation across formal-informal boundaries but does not directly inform whether informal wages anchor on the formal MW.

M Annual Migration of the Bunching Spike

Table A14 reports the annual share of formal dependent workers in the S/25 bin starting at each historical MW level (S/750, S/850, S/930, S/1,025), from 2015 through 2023. The pattern documents the dynamic adjustment described in Section 5.3.

Table A14: Annual Migration of the Bunching Spike across MW Vintages

Year	S/750	S/850	S/930	S/1,025	Notes
2015	5.15	2.79	1.50	0.29	Event A pre
2016	2.36	5.15	1.25	0.20	MW S/750→S/850 (May)
2017	0.56	6.95	1.68	0.23	Event A post; Event B pre
2018	0.41	3.05	5.92	0.34	MW S/850→S/930 (Apr)
2019	0.37	0.53	7.72	0.20	Event B post
2021	0.39	0.35	10.51	0.33	Event C pre
2022	0.28	0.40	3.87	3.09	MW S/930→S/1,025 (May)
2023	0.38	0.26	0.54	7.14	Event C post

Note: Cells report the share, in percent, of formal dependent workers in the S/25 bin beginning at the indicated wage vintage. Sample and wage rule are those of Table 2, so the column for the contemporaneous minimum wage reproduces its share-at-MW row. ENAHO Module 500, employed dependent workers ($p507 \in \{3, 4, 6\}$) who are formal ($ocupinf = 2$), monthly wage in $(0, 6{,}000)$, weighted by *fac500a*. The 2020 survey is excluded. The spike migrates from each old floor to its successor over roughly two years, with the policy-change years showing partial bunching at both vintages.

N Regression and Falsification Output

This appendix tabulates the principal event-study coefficients, the identification diagnostics and the two falsification grids that Sections 5.2, 5.3 and 7.2 refer to. The event-study coefficients are reported one event per table, and each falsification grid is reported one exercise and one event per table, so that no table carries a row whose event or exercise label sits on an earlier page.

Table A15: Event A: Department-Level Event-Study Coefficients, per 10 Percentage Points of Exposure

Outcome	Year	Coefficient	95% CI	Boot p
Employment rate	2013	−1.66	[−3.96, +2.90]	0.390
	2014	−1.79	[−4.46, +1.46]	0.374
	2016	+0.61	[−1.47, +3.38]	0.540
	2017	−0.02	[−1.89, +2.46]	0.980
Formal employment rate	2013	+0.51	[−1.17, +2.50]	0.526
	2014	−0.43	[−4.03, +3.59]	0.757
	2016	+0.46	[−3.19, +7.69]	0.792
	2017	+2.06	[−4.77, +8.09]	0.389
Informal self-employment	2013	−0.43	[−6.45, +4.55]	0.822
	2014	+1.95	[−1.05, +7.15]	0.183
	2016	+2.57	[−2.37, +7.90]	0.209
	2017	−0.21	[−3.75, +5.35]	0.871

Note: Continuous pre-policy exposure interacted with year dummies, base year omitted, department fixed effects, no exposure level term. Coefficients scaled to a 10 percentage point increase in exposure. Intervals and p-values are wild cluster bootstrap-t by test inversion, Rademacher weights, null imposed, 9,999 replications, clustered on department. Individual controls are age, age squared, gender and education. The sample is urban workers aged 18 to 65 with positive survey weights. Joint lead restriction bootstrap p-values for every event and outcome are reported in Table A18.

Table A16: Event B: Department-Level Event-Study Coefficients, per 10 Percentage Points of Exposure

Outcome	Year	Coefficient	95% CI	Boot p
Employment rate	2014	−0.97	[−4.39, +2.57]	0.555
	2015	−0.70	[−2.80, +1.56]	0.506
	2016	+1.26	[−2.43, +4.89]	0.570
	2018	+2.58	[−1.94, +6.65]	0.255
	2019	+0.70	[−3.14, +4.31]	0.672
Formal employment rate	2014	−1.96	[−5.90, +2.76]	0.292
	2015	−1.61	[−5.95, +2.72]	0.478
	2016	+1.00	[−2.23, +4.20]	0.526
	2018	+0.75	[−1.00, +2.73]	0.366
	2019	−1.04	[−2.95, +1.10]	0.285
Informal self-employment	2014	+1.65	[−1.09, +4.08]	0.203
	2015	+0.24	[−3.05, +3.43]	0.867
	2016	+1.74	[−2.94, +5.64]	0.381
	2018	+3.98	[+0.06, +7.16]	0.046
	2019	+2.13	[−1.54, +5.44]	0.243
Informal self-emp., conditional	2014	+0.32	[−3.36, +3.72]	0.847
	2015	+0.16	[−4.80, +4.52]	0.939
	2016	−0.33	[−6.62, +5.48]	0.911
	2018	+4.30	[+0.24, +7.98]	0.035
	2019	+3.80	[−1.09, +7.54]	0.122

Note: Specification, scaling and inference as in Table A15.

Table A17: Event C: Department-Level Event-Study Coefficients, per 10 Percentage Points of Exposure

Outcome	Year	Coefficient	95% CI	Boot p
Employment rate	2017	+6.05	[+1.35, +9.18]	0.003
	2018	+5.86	[+2.69, +7.74]	0.000
	2019	+4.27	[+1.57, +6.68]	0.001
	2022	+3.06	[+1.31, +4.72]	0.010
	2023	+4.67	[+1.70, +7.56]	0.010
Formal employment rate	2017	+1.60	[−1.14, +3.00]	0.219
	2018	+1.43	[−1.33, +2.79]	0.295
	2019	−0.07	[−2.39, +1.31]	0.921
	2022	+0.83	[−0.10, +1.76]	0.066
	2023	+0.55	[−1.35, +1.98]	0.585
Informal self-employment	2017	−0.31	[−2.48, +1.98]	0.705
	2018	+0.78	[−2.21, +3.98]	0.550
	2019	+1.57	[−1.34, +4.55]	0.210
	2022	−0.01	[−2.06, +1.41]	0.991
	2023	−0.39	[−3.13, +2.78]	0.740

Note: Specification, scaling and inference as in Table A15.

Table A18 gathers the identification diagnostics for the department-level design. Panel A tests the design on a window in which the nominal floor did not move, and Panel B tests the lead coefficients within each event window. The diagnostics do not reject for Event B. For Event A, employment and formal-employment leads do not reject, but the informal-self-employment lead restriction rejects at $p = 0.032$. Both employment outcomes reject in Event C, which is why that event is not identified through this design.

Table A18: Department-Level Identification Diagnostics

Test	Window or event	Employment	Formal emp.	Informal SE
Panel A: placebo on the constant-floor window, joint F-test p				
Event A dose	2013–2015	0.830	0.750	—
Wage-range dose	2013–2015	0.78	0.39	0.66
Panel B: joint lead restriction, bootstrap p				
Lead coefficients	Event A	0.638	0.823	0.032
	Event B	0.340	0.575	0.606
	Event C	0.007	0.012	0.070
Conditional on Event A	Event B	—	—	0.991
Panel C: Event B informal self-employment, leave-one-out				
Attenuation excluding Lima	Event B	—	—	59%

Note: Panel A uses the 2013–2015 window, in which the nominal floor was constant at S/750. The Event A dose row uses the actual share-of-treated dose computed from the 2015 affected zone. The wage-range dose row redefines the dose on the 2013 zone $[0.85 \times 750, 750)$. Self-employment is not tested under the Event A dose. Panel B tests the pre-period interaction coefficients jointly by wild cluster bootstrap-t with Rademacher weights and the null imposed, 9,999 replications, clustered on department. The conditional row adds Event A exposure interacted with year dummies and applies only to the informal self-employment outcome. Panel C reports the reduction in the Event B 2018 coefficient when Lima is dropped from the estimation sample. With 25 departments these tests have limited power, so non-rejection is a diagnostic and not a validation of parallel trends.

The four tables that follow report the EPE falsification grid, one exercise and one event each. All entries are in percentage points. Placebo dates are pseudo-events with no minimum wage change inside the estimation span, excluding the March to October 2020 fieldwork disruption. Placebo thresholds hold the event date fixed and move the floor to k times the statutory pair. Baseline labels: matched is the matched-calendar baseline, A1 the CLR-

trend baseline with ridge penalty, and B1 the one-factor clean-bin baseline, all three defined in Section 4.2. Every cell uses a common estimation span of 12 pre-event and 6 post-event rolling windows, so that placebo dates late in the series are estimated on the same span as early ones. Intervals are 95 percent from the unstratified PSU-trajectory bootstrap with 499 replications. Rejects reports whether the 95 percent interval lies entirely above zero, indicating positive accumulation.

Table A19: EPE Falsification, Placebo Dates: Event A

Cell	Δ_E	95% CI	Rejects
2015-01 (matched)	−0.585	[−1.895, +0.722]	no
2015-01 (A1)	+0.242	[−1.313, +2.096]	no
2015-01 (B1)	−0.876	[−2.322, +0.131]	no
2015-03 (matched)	−0.737	[−2.188, +0.638]	no
2015-03 (A1)	+0.703	[−0.775, +2.295]	no
2015-03 (B1)	−0.693	[−2.069, +0.276]	no
2015-05 (matched)	−1.058	[−2.404, +0.236]	no
2015-05 (A1)	+0.149	[−1.501, +1.696]	no
2015-05 (B1)	−0.964	[−2.328, −0.048]	no
2015-07 (matched)	−0.609	[−2.041, +0.868]	no
2015-07 (A1)	−0.158	[−1.663, +1.782]	no
2015-07 (B1)	−0.948	[−2.419, +0.073]	no
2017-04 (matched)	+0.537	[−0.978, +2.122]	no
2017-04 (A1)	−0.802	[−2.698, +1.391]	no
2017-04 (B1)	+0.083	[−1.606, +1.220]	no
2017-06 (matched)	+0.595	[−0.938, +2.052]	no
2017-06 (A1)	−0.416	[−2.280, +1.605]	no
2017-06 (B1)	−0.106	[−1.712, +1.082]	no
2019-03 (matched)	−1.933	[−3.247, −0.502]	no
2019-03 (A1)	−0.547	[−1.985, +1.175]	no
2019-03 (B1)	−1.443	[−2.764, −0.519]	no
2019-05 (matched)	−1.518	[−2.866, −0.158]	no
2019-05 (A1)	+0.396	[−1.096, +2.001]	no
2019-05 (B1)	−1.466	[−2.945, −0.604]	no

Note: Definitions, baselines and inference as described above.

Table A20: EPE Falsification, Placebo Dates: Event B

Cell	Δ_E	95% CI	Rejects
2015-01 (matched)	−0.575	[−1.715, +0.701]	no
2015-01 (A1)	+0.129	[−1.247, +1.622]	no
2015-01 (B1)	−0.487	[−1.676, +0.469]	no
2015-03 (matched)	−0.649	[−1.842, +0.695]	no
2015-03 (A1)	−0.322	[−1.679, +1.154]	no
2015-03 (B1)	−0.975	[−2.159, +0.121]	no
2015-05 (matched)	−0.480	[−1.628, +0.839]	no
2015-05 (A1)	−0.704	[−2.005, +0.723]	no
2015-05 (B1)	−1.066	[−2.241, −0.130]	no
2015-07 (matched)	−0.117	[−1.421, +1.080]	no
2015-07 (A1)	−0.278	[−1.710, +1.291]	no
2015-07 (B1)	−0.455	[−1.887, +0.451]	no
2017-04 (matched)	−0.046	[−1.254, +1.170]	no
2017-04 (A1)	−0.887	[−2.299, +0.836]	no
2017-04 (B1)	−0.593	[−1.917, +0.379]	no
2017-06 (matched)	−0.085	[−1.240, +1.100]	no
2017-06 (A1)	−0.846	[−2.255, +0.683]	no
2017-06 (B1)	−0.420	[−1.602, +0.445]	no
2019-03 (matched)	+0.059	[−1.188, +1.489]	no
2019-03 (A1)	−0.661	[−2.254, +1.068]	no
2019-03 (B1)	+0.419	[−0.766, +1.616]	no
2019-05 (matched)	−0.067	[−1.392, +1.211]	no
2019-05 (A1)	+0.280	[−1.247, +1.827]	no
2019-05 (B1)	−0.411	[−1.727, +0.422]	no

Note: Definitions, baselines and inference as described above.

Table A21: EPE Falsification, Placebo Thresholds: Event A

Cell	Δ_E	95% CI	Rejects
statutory (matched)	+2.113	[+0.661, +3.662]	yes
statutory (A1)	+2.904	[+1.158, +4.668]	yes
statutory (B1)	+1.887	[+0.321, +3.054]	yes
$k = 0.60$ (matched)	−0.424	[−1.157, +0.449]	no
$k = 0.60$ (A1)	+0.800	[−0.107, +1.738]	no
$k = 0.60$ (B1)	−0.090	[−0.747, +0.747]	no
$k = 0.75$ (matched)	−2.683	[−3.969, −1.314]	no
$k = 0.75$ (A1)	−0.977	[−2.462, +0.526]	no
$k = 0.75$ (B1)	−2.542	[−3.827, −1.511]	no
$k = 0.90$ (matched)	−0.164	[−1.681, +1.333]	no
$k = 0.90$ (A1)	+1.105	[−0.713, +2.961]	no
$k = 0.90$ (B1)	−0.319	[−1.929, +0.839]	no
$k = 1.15$ (matched)	−0.427	[−1.631, +0.991]	no
$k = 1.15$ (A1)	−0.883	[−2.723, +0.969]	no
$k = 1.15$ (B1)	−1.114	[−2.555, −0.008]	no
$k = 1.30$ (matched)	+0.154	[−1.076, +1.490]	no
$k = 1.30$ (A1)	−0.484	[−2.134, +1.081]	no
$k = 1.30$ (B1)	−0.521	[−1.886, +0.484]	no
$k = 1.50$ (matched)	+0.302	[−0.905, +1.722]	no
$k = 1.50$ (A1)	+0.101	[−1.233, +1.761]	no
$k = 1.50$ (B1)	−0.200	[−1.405, +0.914]	no

Note: Definitions, baselines and inference as described above. The statutory rows are not expected to reproduce Table 7, which uses the full post-event span of 11 windows and 999 replications. This grid holds the span fixed at 6 post windows for comparability, so its statutory estimates are the same quantity measured over a shorter horizon.

Table A22: EPE Falsification, Placebo Thresholds: Event B

Cell	Δ_E	95% CI	Rejects
statutory (matched)	+4.754	[+3.450, +6.041]	yes
statutory (A1)	+4.628	[+3.349, +6.299]	yes
statutory (B1)	+4.315	[+2.933, +5.237]	yes
$k = 0.60$ (matched)	−0.914	[−1.635, −0.127]	no
$k = 0.60$ (A1)	+0.255	[−0.546, +1.105]	no
$k = 0.60$ (B1)	−0.386	[−1.008, +0.253]	no
$k = 0.75$ (matched)	−1.816	[−3.015, −0.621]	no
$k = 0.75$ (A1)	−1.507	[−2.718, −0.186]	no
$k = 0.75$ (B1)	−2.079	[−3.198, −1.299]	no
$k = 0.90$ (matched)	−1.315	[−2.723, +0.007]	no
$k = 0.90$ (A1)	−1.402	[−2.872, +0.253]	no
$k = 0.90$ (B1)	−1.841	[−3.365, −1.012]	no
$k = 1.15$ (matched)	+1.962	[+0.544, +3.211]	yes
$k = 1.15$ (A1)	+1.614	[+0.107, +3.197]	yes
$k = 1.15$ (B1)	+0.998	[−0.522, +1.828]	no
$k = 1.30$ (matched)	+0.916	[−0.273, +2.009]	no
$k = 1.30$ (A1)	+0.978	[−0.183, +2.391]	no
$k = 1.30$ (B1)	+0.266	[−0.948, +0.983]	no
$k = 1.50$ (matched)	+0.201	[−0.896, +1.189]	no
$k = 1.50$ (A1)	+0.052	[−1.084, +1.406]	no
$k = 1.50$ (B1)	−0.335	[−1.363, +0.384]	no

Note: Definitions, baselines and inference as described above. The statutory rows are not expected to reproduce Table 7, which uses the full post-event span of 11 windows and 999 replications. This grid holds the span fixed at 6 post windows for comparability, so its statutory estimates are the same quantity measured over a shorter horizon.

The corresponding grid for the 2025 increase follows, again split by exercise. It uses the

national quarterly series, five pseudo-event dates instead of eight, and six placebo thresholds on the same k grid.

Table A23: Event D Falsification, Placebo Dates: Constant Floor at S/1,025

Cell	Baseline	Δ_E	95% CI	Rejects
2023-07	Matched calendar	−1.15	[−1.90, −0.42]	no
2023-07	CLR trend	−1.12	[−1.88, −0.64]	no
2023-07	Clean-bin factor	−1.15	[−1.90, −0.65]	no
2023-10	Matched calendar	−0.26	[−1.02, +0.51]	no
2023-10	CLR trend	−0.14	[−0.89, +0.33]	no
2023-10	Clean-bin factor	−0.24	[−0.98, +0.25]	no
2024-01	Matched calendar	−0.41	[−1.23, +0.35]	no
2024-01	CLR trend	−0.14	[−0.92, +0.36]	no
2024-01	Clean-bin factor	−0.27	[−1.05, +0.25]	no
2024-04	Matched calendar	−0.30	[−1.01, +0.41]	no
2024-04	CLR trend	−0.84	[−1.57, −0.38]	no
2024-04	Clean-bin factor	−0.92	[−1.63, −0.48]	no
2024-07	Matched calendar	−0.40	[−1.09, +0.26]	no
2024-07	CLR trend	−1.03	[−1.79, −0.57]	no
2024-07	Clean-bin factor	−1.17	[−1.92, −0.76]	no

Note: Percentage points, national quarterly series, S/10 bins. Placebo dates are pseudo-events inside the constant-floor window on S/1,025, each estimated on a fixed two-quarter post span so that early and late pseudo-events are comparable. Intervals are 95 percent from the stratified PSU-trajectory bootstrap with 999 replications. Rejects reports whether the interval lies entirely above zero.

Table A24: Event D Falsification, Placebo Thresholds: Date Fixed at 2025Q1

Cell	Baseline	Δ_E	95% CI	Rejects
statutory	Matched calendar	+2.46	[+1.94, +2.97]	yes
statutory	CLR trend	+2.73	[+2.10, +3.14]	yes
statutory	Clean-bin factor	+2.06	[+1.47, +2.40]	yes
$k = 0.60$	Matched calendar	−0.59	[−0.96, −0.23]	no
$k = 0.60$	CLR trend	−0.18	[−0.52, +0.13]	no
$k = 0.60$	Clean-bin factor	−1.15	[−1.44, −0.81]	no
$k = 0.75$	Matched calendar	−4.72	[−5.24, −4.21]	no
$k = 0.75$	CLR trend	−4.62	[−5.23, −4.30]	no
$k = 0.75$	Clean-bin factor	−5.77	[−6.31, −5.55]	no
$k = 0.90$	Matched calendar	−1.88	[−2.43, −1.31]	no
$k = 0.90$	CLR trend	−1.95	[−2.70, −1.53]	no
$k = 0.90$	Clean-bin factor	−2.66	[−3.39, −2.43]	no
$k = 1.15$	Matched calendar	+0.16	[−0.36, +0.67]	no
$k = 1.15$	CLR trend	−0.08	[−0.70, +0.30]	no
$k = 1.15$	Clean-bin factor	+0.18	[−0.41, +0.44]	no
$k = 1.30$	Matched calendar	+0.22	[−0.34, +0.69]	no
$k = 1.30$	CLR trend	+0.06	[−0.53, +0.42]	no
$k = 1.30$	Clean-bin factor	+0.43	[−0.11, +0.70]	no
$k = 1.50$	Matched calendar	+0.41	[−0.08, +0.86]	no
$k = 1.50$	CLR trend	+0.35	[−0.13, +0.75]	no
$k = 1.50$	Clean-bin factor	+0.47	[+0.09, +0.81]	yes

Note: Placebo thresholds hold the date at 2025Q1 and scale both floors by k. The statutory row uses the full five-quarter post span and therefore does not match the two-quarter placebo-date convention of Table A23. Series, bins and inference as in that table.

O Variable Mapping

Table A25 maps the conceptual variables used in the analysis to their exact survey identifiers, for replication purposes. All entries are ENAHO Module 500 variables except the panel expansion factor, which belongs to the Module 978 rotating panel, and the EsSalud registration marker, which belongs to EPE Lima. The table names the source for both. Variable codes follow the INEI 2021 questionnaire conventions.

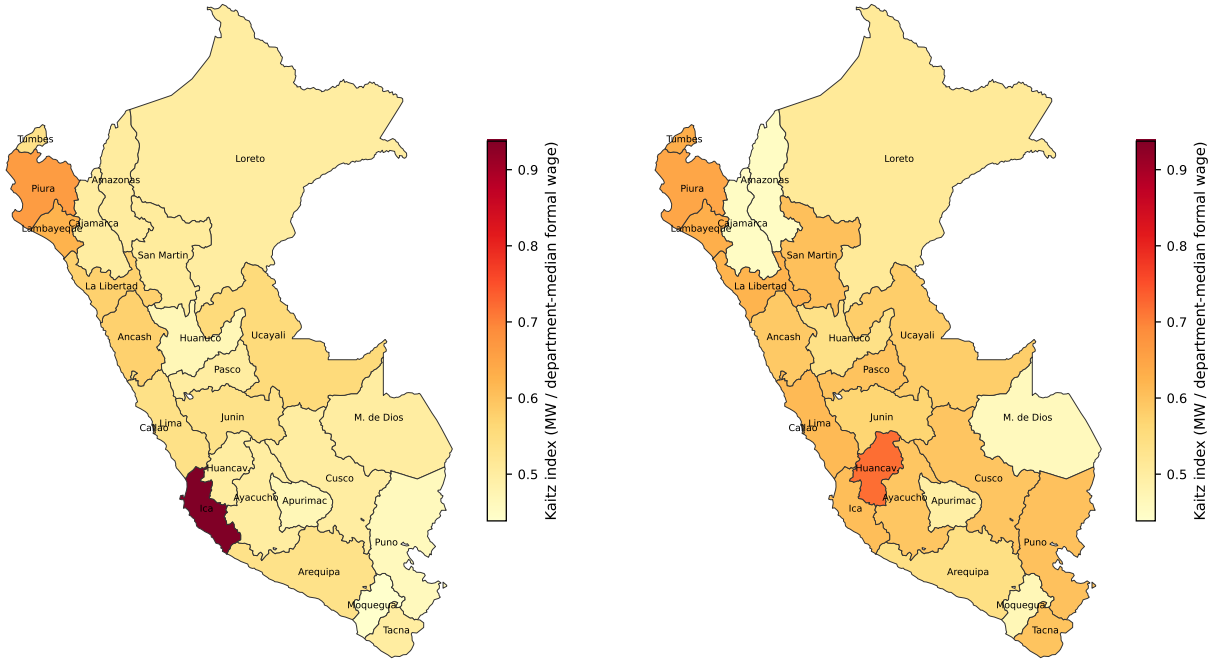
Table A25: Survey Variables Used in the Analysis

Concept	Variable	Definition
Employment status	ocu500	=1 if employed in reference week
Worker type	p507	=3 empleado, =4 obrero, =6 trabajador del hogar
Self-employment	p507	=2 trabajador independiente
Formality	ocupinf	=2 if formal (INEI definition)
Monthly wage	p524a1	Cash income from primary occupation
Imputed wage	i524a1	Imputed annual income / 12 (where p524a1 is not positive)
Self-employment income	i530a	Annual net business income, divided by twelve
Weekly hours	p513t	Total weekly hours worked
Age	p208a	Years of age (alt. *edad*)
Sex	p207	=1 male (alt. *sexo*)
Education	p301a	Highest level achieved (alt. *p301*)
Industry	p506r4	CIIU 2-digit sector code
Firm size	p512b	Establishment worker count
Employer type	p510	=1,2,3 for central, regional, local government
Geographic identifier	ubigeo	6-digit, first 2 = department
Urban/rural stratum	estrato	≤ 5 urban, > 5 rural
Survey expansion factor	fac500a	Person-level weight for Module 500
Panel expansion factor	facpanel2123	For 2021–2023 panel (Module 978, not 500)
EsSalud registration	p222	=1 if registered, public health insurer (EPE Lima)

Note: Variables p208a, p207, p301a may appear with alternative names (*edad*, *sexo*, *p301*) across ENAHO releases. These alternative variable names are harmonized to the concepts listed in the table. The primary bunching sample restricts to dependent workers ($p507 \in \{3, 4, 6\}$). The informal self-employment outcome used in the regression composition-shift analysis (Section 7.2) and in the composition decomposition (Section 7.1) combines $p507 = 2$ with $ocupinf = 1$, so it is the informal category and not total self-employment. The p222 variable is from EPE Lima (Encuesta Permanente de Empleo) and is used only as the EsSalud contributory marker in the high-frequency analysis of Section 5.3.

P Department Kaitz Rankings

Figure A4 compares the department-level Kaitz index between the Event A pre-period (2015, MW = S/750) and the Event C pre-period (2021, MW = S/930). The Kaitz index is the ratio of the nominal minimum wage to the weighted-median formal monthly wage in each department. Higher (darker) values indicate the minimum wage is more binding relative to the local formal wage distribution. Both panels use ENAHO Module 500 formal dependent workers and a common color scale to make the spatial reshuffling directly visible.



(a) 2015, Event A pre-period (MW = S/750)

(b) 2021, Event C pre-period (MW = S/930)

Figure A4: Department Kaitz Rankings, 2015 and 2021

Note: Left panel uses 2015 ENAHO Module 500 data, MW = S/750. Right panel uses 2021 ENAHO Module 500 data, MW = S/930. Kaitz = MW divided by department-level weighted median monthly wage of formal dependent workers. Huancavelica and Madre de Dios are abbreviated to Huancav. and M. de Dios so that the labels fit inside their polygons. Table A26 lists every department in full.

Table A26 reports the 2015 and 2021 Kaitz values by department and the change be-

tween events. Two departments stand out. Ica falls from 0.94 in 2015 (the highest in the country) to 0.61 in 2021 as agro-export formal wages roughly doubled in nominal terms over the period, driven by the expansion of asparagus, table-grape, and avocado production. Huancavelica rises from 0.50 to 0.72 as the formal wage median in the department fell from approximately S/1,500 to S/1,292 over the same period, plausibly reflecting compositional shifts away from higher-paid public-sector formal employment toward lower-paid formal alternatives. Cajamarca, Amazonas, and Madre de Dios show the lowest 2021 values (0.45 to 0.46), supported by formal wages in mining and large-scale agriculture that exceed the MW comfortably. The substantial inter-event reshuffling of the Kaitz ranking is the reason the share-of-treated DiD in Section 7.2 uses department-specific shares of workers in the affected wage range as a treatment-intensity measure in place of the Kaitz index itself.

Table A26: Kaitz Index by Department, 2015 vs 2021

Department	2015	2021	Change	Department	2015	2021	Change
Amazonas	0.50	0.45	−0.05	Lambayeque	0.63	0.63	+0.01
Ancash	0.58	0.59	+0.01	Lima	0.54	0.61	+0.07
Apurimac	0.47	0.49	+0.03	Loreto	0.50	0.51	+0.01
Arequipa	0.54	0.54	+0.01	Madre de Dios	0.50	0.46	−0.04
Ayacucho	0.50	0.59	+0.09	Moquegua	0.44	0.47	+0.03
Cajamarca	0.50	0.45	−0.05	Pasco	0.50	0.60	+0.10
Callao	0.63	0.63	+0.00	Piura	0.66	0.65	−0.02
Cusco	0.50	0.60	+0.10	Puno	0.46	0.60	+0.14
Huancavelica	0.50	0.72	+0.22	San Martin	0.51	0.60	+0.10
Huanuco	0.47	0.54	+0.07	Tacna	0.50	0.60	+0.10
Ica	0.94	0.61	−0.33	Tumbes	0.54	0.63	+0.10
Junin	0.54	0.57	+0.03	Ucayali	0.56	0.58	+0.03
La Libertad	0.58	0.62	+0.05				

Note: Weighted median formal monthly wage by department, computed on ENAHO Module 500 with formal dependent workers (*empleados*, *obreros*, *trabajadores del hogar*). Kaitz 2015 = S/750 / median 2015. Kaitz 2021 = S/930 / median 2021. Derived columns are computed from unrounded inputs, so they may differ in the last digit from the difference of the displayed rounded values.